

ASVAB Math Workbook

1,000 Practice Problems with Step-by-Step Solutions

Arithmetic Reasoning & Mathematics Knowledge

Key concepts and worked examples for every topic

A diagnostic test with a personal study plan

25 drill problems in 1 chapters

4 full-length math practice tests

Every answer checked by computer algebra

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ASVAB Math Workbook: 1,000 Practice Problems with Step-by-Step Solutions

First edition, 2026

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Test formats, question counts, time limits, and qualifying scores are set by the Department of Defense and the individual services and can change. Always confirm current details with your recruiter or at officialasvab.com. The author makes no guarantee of any particular test score.

The practice problems in this book were generated and their answers verified with the SymPy computer algebra system; every answer was computed exactly and then confirmed by a second, independent calculation.

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How to Use This Book

The math on the ASVAB is mostly middle-school and early high-school math—percents, fractions, equations, areas, word problems—asked quickly and without a calculator. What raises scores is *practice*: seeing each kind of question often enough that you recognize it instantly and know the first step.

This workbook gives you that practice: **1,000 problems**, each with a complete step-by-step solution.

The plan

1. **Take the Diagnostic Test** (page 5). Thirty questions, about 35 minutes, no calculator. The answer key tells you which chapter each question comes from.
2. **Study the chapters you need.** Start with the topics you missed on the diagnostic. Each chapter has three parts:
 - *Key Concepts*—the rules, formulas, a worked example, a no-calculator tip, and the traps test writers love.
 - *Practice Set*—warm-up problems first, then test-level problems, then challenge problems.
 - *Answers & Explanations*—a quick answer key plus a full solution for every problem.
3. **Take the four Practice Tests** under real time limits once you have worked through the chapters. Each has an Arithmetic Reasoning section (30 questions, 36 minutes) and a Mathematics Knowledge section (25 questions, 24 minutes), the format of the paper ASVAB.
4. **Review every miss.** The answer keys of the tests list the chapter for every question, so you always know where to go back.

Reading the solutions

Every solution is broken into numbered steps that you can follow with pencil and paper. Many solutions also include:

- **TIP**—a shortcut, an estimate, or a way to check your answer.
- **WHY NOT**—what went wrong if you picked one of the tempting wrong answers. The wrong choices in this book are not random numbers; each comes from a real, common mistake (dividing by the wrong number, forgetting to convert minutes to hours, adding percents that should be multiplied). If your answer matches one of them, the explanation tells you exactly which mistake you made.

Good habits

- **No calculator.** The ASVAB does not allow one, so don't practice with one. Every problem here is designed to be done by hand.
- **Write your work down.** Scratch paper is provided at the test; use it here too.
- **Time yourself.** Each practice set shows a target time. Don't worry about speed at first—accuracy comes first, speed follows.
- **Redo your misses a week later** and log them in the Progress Tracker at the back of the book.

The ASVAB Math Subtests

The Armed Services Vocational Aptitude Battery (ASVAB) is the entrance test for the U.S. Army, Navy, Air Force, Marine Corps, Space Force, and Coast Guard. It consists of several subtests. Two of them are math:

Arithmetic Reasoning (AR)
Word problems. Each question describes a real-life situation—a purchase, a trip, a paycheck, a recipe, a work schedule—and you have to decide which operations to use. Typical topics: percents and discounts, ratios, rates and distance, averages, interest, work, measurement, and multi-step reasoning.
Mathematics Knowledge (MK)
Direct math questions: number properties, fractions, exponents and roots, algebra (expressions, equations, inequalities, factoring), and geometry (angles, triangles, area, circles, volume, coordinates). Knowing the vocabulary and formulas is half the battle.

Format and timing

The ASVAB is given in two main formats. On the **computer-adaptive** version (CAT-ASVAB), the difficulty of each question depends on how you answered the previous ones, and you cannot go back to change an answer. On the **paper-and-pencil** version, everyone answers the same questions and you can move around within a subtest. Typical question counts and time limits:¹

	Arithmetic Reasoning	Mathematics Knowledge
CAT-ASVAB (computer)	15 questions, 55 minutes	15 questions, 31 minutes
Paper and pencil	30 questions, 36 minutes	25 questions, 24 minutes

Every question is multiple choice with four answer choices. **Calculators are not allowed**; scratch paper is provided. Formulas are *not* given, so memorize the ones on the Formula Sheet at the back of this book.

Why math matters so much

Your Armed Forces Qualification Test (AFQT) score—the score that decides whether you can enlist at all—is computed from four subtests: Arithmetic Reasoning, Mathematics Knowledge, Word Knowledge, and Paragraph Comprehension. The two math subtests make up roughly **half** of the AFQT. The AFQT is reported as a percentile from 1 to 99, and each service sets its own minimum.

Arithmetic Reasoning and Mathematics Knowledge also feed many of the *line scores* (composite scores) that qualify you for specific jobs, such as technical, electronics, and administrative specialties. A few extra correct math answers can open up a wider choice of careers and enlistment bonuses.

¹These figures can change; confirm them with your recruiter or at officialasvab.com.

Test-Day Strategies

1. Read the last sentence first

In a word problem, the last sentence tells you what to find. Read it first, then read the whole problem looking for the numbers you need. Many wrong answer choices are the right answer to a *different* question (the discount instead of the sale price, the number who passed instead of the number who failed).

2. Estimate before you calculate

A rough answer tells you which choices are impossible. If a \$48 item is 25% off, the sale price must be a bit more than \$35—so \$12 and \$60 are out before you do any exact arithmetic.

3. Work backward from the choices

When a question asks “which number...” or gives an equation, you can test the answer choices. Start with a middle value; if it is too big, the answer is smaller, and vice versa. For “which value satisfies” questions this is often the fastest method.

4. Pick numbers

If the problem uses variables or percents of an unknown amount, choose an easy number (100 for percents, a common multiple for fractions) and see which answer choice gives the same result.

5. Watch the units

Minutes vs. hours, inches vs. feet, cents vs. dollars, radius vs. diameter. Convert everything to the units the question asks for *before* you calculate.

6. Manage your time

On the paper test you have a little over a minute per question. If a question takes more than two minutes, make your best guess, mark it, and move on. Never leave a question blank: a blank is always wrong, a guess might be right, and on the computer test unanswered questions at the end of a subtest lower your score.

7. Check that your answer makes sense

A sale price cannot be higher than the original price. A probability cannot be more than 1. An average must lie between the smallest and largest values. A quick reasonableness check catches many careless errors.

No-Calculator Tip Learn by heart: the squares from 1^2 to 15^2 , the fraction–decimal–percent equivalents ($\frac{1}{8} = 0.125 = 12.5\%$, $\frac{1}{3} \approx 33.3\%$, ...), and the formulas on the Formula Sheet. Instant recall of these saves minutes on test day.

Mental Math Toolkit

Because the ASVAB bans calculators, a handful of shortcuts pay off on almost every page of this book. Practice them until they are automatic.

Trick	Example
Multiply by 5	Multiply by 10, then halve: $48 \times 5 = 480 \div 2 = 240$.
Multiply by 25	Multiply by 100, then divide by 4: $36 \times 25 = 3,600 \div 4 = 900$.
Multiply by 9 or 11	$47 \times 9 = 470 - 47 = 423$; $34 \times 11 = 340 + 34 = 374$.
Break a number apart	$7 \times 98 = 7 \times 100 - 7 \times 2 = 700 - 14 = 686$.
Square a number ending in 5	35^2 : $3 \times 4 = 12$, then write 25: 1,225.
Divide by 5	Double, then divide by 10: $340 \div 5 = 680 \div 10 = 68$.
Divide by a decimal	Move both decimal points: $4.8 \div 0.06 = 480 \div 6 = 80$.
Build percents from 10%	15% of 60: 10% = 6, 5% = 3, total 9.
Swap the percent	8% of 50 = 50% of 8 = 4.
Compare fractions	Cross-multiply: $\frac{5}{8}$ vs. $\frac{3}{5}$: $5 \times 5 = 25 > 8 \times 3 = 24$, so $\frac{5}{8}$ is larger.
Estimate first	$398 \times 51 \approx 400 \times 50 = 20,000$ rules out answers near 2,000 or 200,000.

Diagnostic Test

This test samples every chapter of the book. Take it before you begin studying, without a calculator, and try to finish in about 1 minutes. Don't guess wildly—if you have no idea how to start a problem, leave it blank so that the result shows what you really need to study.

Score: ____ / 1

Target time: 1 minutes

1. Which of the following is equal to 0.86?
A 86% B 106% C 126% D 860%

Answers and Your Study Plan

Check your answers against the table. Each row shows the chapter the question comes from. **Circle the chapter number of every question you missed or left blank**—those chapters are your study plan. Work through them first, then the rest of the book in order.

#	Ans.	Ch.	Topic	#	Ans.	Ch.	Topic
1	A	6	Percents				

How did you do?

26–30 correct	Strong. Skim the Key Concepts, do the challenge problems, then the practice tests.
18–25 correct	Solid base. Study your circled chapters fully, then the rest.
10–17 correct	Work through every chapter in order; take your time with the warm-ups.
0–9 correct	Start at Chapter 1. Every chapter builds on the ones before it.

Explanations

1. **A** 86%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.86 \times 100 = 86$, so $0.86 = 86\%$.

WHY NOT (D) moves the decimal point three places.

PART I

Number Skills

Whole numbers, signed numbers, factors, fractions, decimals, percents, ratios, exponents, and roots—the toolkit behind every other question.

Percents

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Percent means “per hundred.” Percent questions appear on both math subtests, and most of them reduce to one sentence: **part = percent × whole**.

Converting between forms

- Percent → decimal: move the decimal point two places *left* ($35\% = 0.35$, $4\% = 0.04$, $150\% = 1.5$).
- Decimal → percent: move it two places *right* ($0.6 = 60\%$, $0.075 = 7.5\%$).
- Fraction → percent: divide, then convert ($\frac{3}{8} = 0.375 = 37.5\%$).

The three percent questions

Every basic percent problem gives you two of the three quantities in

$$\text{part} = \text{percent} \times \text{whole}$$

and asks for the third.

- **Find the part:** 25% of 64 is $0.25 \times 64 = 16$.
- **Find the percent:** 18 is what percent of 72? $\frac{18}{72} = 0.25 = 25\%$.
- **Find the whole:** 24 is 30% of what number? $24 \div 0.30 = 80$.

Percent change

$$\text{percent change} = \frac{\text{new} - \text{original}}{\text{original}} \times 100\%$$

Always divide by the *original* (starting) value.

WORKED EXAMPLE

A jacket costs \$60. Its price goes up 15%. What is the new price?

Solution. The increase is $0.15 \times 60 = 9$ dollars, so the new price is $60 + 9 = \$69$. *Shortcut:* a 15% increase means you pay 115%, and $1.15 \times 60 = 69$.

No-Calculator Tip Benchmark percents make mental math fast: 10% = move the decimal one place left, 5% = half of 10%, 1% = move it two places, 25% = $\frac{1}{4}$, 50% = $\frac{1}{2}$, 75% = $\frac{3}{4}$, 20% = $\frac{1}{5}$. So 35% of 80 is $30\% + 5\% = 24 + 4 = 28$.

Common Traps

- Answering the wrong question: if 35% *passed*, the number who *did not* pass uses 65%.
- Dividing by the new value in a percent-change problem.
- Adding successive percents: up 25% then down 20% is *not* a 5% increase.

Practice Set 6

Score: ____ / 25

Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is 25% of 396?
A 99 B 297 C 495 D 990
- Which of the following is equal to 0.13?
A 1.3% B 13% C 15% D 130%
- What is 75% of 128?
A 32 B 96 C 224 D 960
- What is $\frac{1}{4}$ written as a percent?
A 1% B 2.5% C 25% D 400%
- Of the 380 students at a high school who took the ASVAB, 75% scored above 50 on the AFQT. How many of these students did *not* score above 50?
A 95 B 105 C 285 D 305
- Which of the following is equal to 0.075?
A 7.5% B 8% C 15% D 75%
- A recruiting office processed 364 applications last month. Of these, 75% were incomplete. How many applications were complete?
A 91 B 101 C 273 D 289
- What is 15% of 320?
A 48 B 272 C 368 D 480
- What percent of 364 is 455?
A 1.25% B 12.5% C 80% D 125%
- 88 is 40% of what number?
A 22 B 128 C 220 D 3,520
- What percent of 340 is 255?
A 0.75% B 7.5% C 75% D 115%
- 126 is 60% of what number?
A 21 B 210 C 420 D 7,560
- Last year the price of a gallon of paint was \$20. This year it is \$32. What is the percent increase?
A 12% B 37.5% C 60% D 160%
- If 60% of a number is 168, what is the number?
A 28 B 180 C 228 D 280
- Last year the cost of a yearly gym membership was \$576. This year it is \$720. What is the percent increase?
A 12.5% B 20% C 25% D 144%
- 96 is what percent of 64?
A 1.5% B 15% C 70% D 150%
- Which of the following is equal to 0.015?
A 1.5% B 2% C 3% D 15%
- What is 150% of 104?
A 136 B 156 C 260 D 1,560
- A camping tent was priced at \$152. The store raised the price by 50%, and a month later it marked the new price down by 50%. What is the final price?
A \$114 B \$144 C \$152 D \$228
- What is 10% of 50% of 840?
A 42 B 84 C 420 D 504

21. A store sold a power drill for \$156. It increased the price by 25%, then put the power drill on sale at 20% off the new price. What is the sale price?
A \$156.00 B \$163.80 C \$195.00 D \$312.00
22. A training battalion has 200 recruits. 20% of them are assigned to the morning shift, and $\frac{2}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 16 B 24 C 40 D 80
23. A store sold a pair of running shoes for \$184. It increased the price by 25%, then put the pair of running shoes on sale at 50% off the new price. What is the sale price?
A \$92 B \$115 C \$138 D \$230
24. A training battalion has 1,000 recruits. 30% of them are assigned to the morning shift, and $\frac{1}{3}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 10 B 60 C 100 D 300
25. What is 75% of 40% of 680?
A 102 B 204 C 272 D 782

Answers & Explanations 6

1 A	6 A	11 C	16 D	21 A
2 B	7 A	12 B	17 A	22 A
3 B	8 A	13 C	18 B	23 B
4 C	9 D	14 D	19 A	24 C
5 A	10 C	15 C	20 A	25 B

1. **A** 99

1 Change the percent to a decimal: $25\% = 0.25$.

2 Multiply by the number: $0.25 \times 396 = 99$.

WHY NOT (B) is what is left after removing 25%, not 25% of the number. (C) adds 25% to the number. (D) moves the decimal point only one place.

2. **B** 13%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.13 \times 100 = 13$, so $0.13 = 13\%$.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.

3. **B** 96

1 Change the percent to a decimal: $75\% = 0.75$.

2 Multiply by the number: $0.75 \times 128 = 96$.

WHY NOT (A) is what is left after removing 75%, not 75% of the number. (C) adds 75% to the number. (D) moves the decimal point only one place.

4. **C** 25%

1 Divide the numerator by the denominator: $1 \div 4 = 0.25$.

2 Multiply by 100 to get a percent: $0.25 = 25\%$.

TIP Or scale the fraction to hundredths: $\frac{1}{4} = \frac{25}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place. (D) divides the denominator by the numerator.

5. **A** 95

1 The question asks about the *other* group: $100\% - 75\% = 25\%$.

2 Find that part: $0.25 \times 380 = 95$.

TIP Or find 75% first (285) and subtract: $380 - 285 = 95$.

WHY NOT (C) is the number in the group described, not the number asked for. (D) subtracts the percent as if it were a count.

6. **A** 7.5%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.075 \times 100 = 7.5$, so $0.075 = 7.5\%$.

WHY NOT (D) moves the decimal point three places.

7. **A** 91

1 The question asks about the *other* group: $100\% - 75\% = 25\%$.

2 Find that part: $0.25 \times 364 = 91$.

TIP Or find 75% first (273) and subtract: $364 - 273 = 91$.

WHY NOT (C) is the number in the group described, not the number asked for. (D) subtracts the percent as if it were a count.

8. **A** 48

1 Change the percent to a decimal: $15\% = 0.15$.

2 Multiply by the number: $0.15 \times 320 = 48$.

WHY NOT (B) is what is left after removing 15%, not 15% of the number. (C) adds 15% to the number. (D) moves the decimal point only one place.

9. **D** 125%

1 Write the part over the whole: $\frac{455}{364}$.

2 Simplify and convert: $\frac{455}{364} = 1.25 = 125\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place. (C) divides the whole by the part instead of the part by the whole.

10. **C** 220

1 Use part = percent $\times$ whole: $88 = 0.4 \times w$.

2 Divide both sides by 0.4: $w = 88 \div 0.4 = 220$.

3 Check: 40% of 220 is 88. ✓

WHY NOT (A) slips the decimal point. (B) adds the percent to the part. (D) multiplies by the percent without converting it.

11. **C** 75%

1 Write the part over the whole: $\frac{255}{340}$.

2 Simplify and convert: $\frac{255}{340} = 0.75 = 75\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

12. **B** 210

1 Use part = percent $\times$ whole: $126 = 0.6 \times w$.

2 Divide both sides by 0.6: $w = 126 \div 0.6 = 210$.

3 Check: 60% of 210 is 126. ✓

WHY NOT (A) slips the decimal point. (D) multiplies by the percent without converting it.

13. **C** 60%

- Find the amount of increase: $32 - 20 = 12$ dollars.
- Divide by the *original* price: $\frac{12}{20} = 0.6$.
- Convert to a percent: $0.6 = 60\%$ increase.

WHY NOT (A) gives the dollar change instead of the percent change. (B) divides by the new price instead of the original price. (D) gives the new price as a percent of the old price.

14. **D** 280

- Use part = percent $\times$ whole: $168 = 0.6 \times w$.
- Divide both sides by 0.6: $w = 168 \div 0.6 = 280$.
- Check: 60% of 280 is 168. ✓

WHY NOT (A) slips the decimal point. (C) adds the percent to the part.

15. **C** 25%

- Find the amount of increase: $720 - 576 = 144$ dollars.
- Divide by the *original* price: $\frac{144}{576} = 0.25$.
- Convert to a percent: $0.25 = 25\%$ increase.

WHY NOT (B) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.

16. **D** 150%

- Write the part over the whole: $\frac{96}{64}$.
- Simplify and convert: $\frac{96}{64} = 1.5 = 150\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

17. **A** 1.5%

- To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- $0.015 \times 100 = 1.5$, so $0.015 = 1.5\%$.

WHY NOT (D) moves the decimal point three places.

18. **B** 156

- Change the percent to a decimal: $150\% = 1.5$.
- Multiply by the number: $1.5 \times 104 = 156$.

WHY NOT (C) adds 150% to the number. (D) moves the decimal point only one place.

19. **A** \$114

- Raise the price by 50%: $152 \times 1.5 = 228$.
- Take 50% off the *new* price: $228 \times 0.5 = 114$.
- The final price is \$114.

WHY NOT (C) assumes the two changes cancel out. (D) stops after the increase.

20. **A** 42

- Work from the inside out. First, 50% of 840: $0.5 \times 840 = 420$.
- Then 10% of that result: $0.1 \times 420 = 42$.

TIP Percents of percents multiply: $0.1 \times 0.5 = 0.05$, and $0.05 \times 840 = 42$.

WHY NOT (B) takes only 10% of 840. (C) stops after taking 50%. (D) adds the two percents.

21. **A** \$156.00

- Raise the price by 25%: $156 \times 1.25 = 195$.
- Take 20% off the *new* price: $195 \times 0.8 = 156$.
- The final price is \$156.

WHY NOT (B) combines the changes into a single 5% increase. (C) stops after the increase.

22. **A** 16

- Find the first group: $0.2 \times 200 = 40$.
- Take $\frac{2}{5}$ of that group: $\frac{2}{5} \times 40 = 16$.

WHY NOT (B) is the part of the group that is *not* included. (C) is the size of the first group only. (D) takes $\frac{2}{5}$ of the whole total.

23. **B** \$115

- Raise the price by 25%: $184 \times 1.25 = 230$.
- Take 50% off the *new* price: $230 \times 0.5 = 115$.
- The final price is \$115.

WHY NOT (A) applies only the decrease. (C) combines the changes into a single 25% decrease. (D) stops after the increase.

24. **C** 100

- Find the first group: $0.3 \times 1,000 = 300$.
- Take $\frac{1}{3}$ of that group: $\frac{1}{3} \times 300 = 100$.

WHY NOT (D) is the size of the first group only.

25. **B** 204

- Work from the inside out. First, 40% of 680: $0.4 \times 680 = 272$.
- Then 75% of that result: $0.75 \times 272 = 204$.

TIP Percents of percents multiply: $0.75 \times 0.4 = 0.3$, and $0.3 \times 680 = 204$.

WHY NOT (C) stops after taking 40%. (D) adds the two percents.

PART II

Practice Tests

Four full-length math sections in the paper-and-pencil format:
30 Arithmetic Reasoning questions in 36 minutes and 25
Mathematics Knowledge questions in 24 minutes.

Practice Test 1

Practice Test 2

Practice Test 3

Practice Test 4

Practice Test 1

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

1. Last year monthly rent for an apartment was \$1,900. This year it is \$1,140. What is the percent decrease?
A 20% B 40% C 60% D 760%
2. Last year monthly rent for an apartment was \$1,500. This year it is \$1,125. What is the percent decrease?
A 25% B 30% C 75% D 375%
3. Last year monthly rent for an apartment was \$1,300. This year it is \$650. What is the percent decrease?
A 50% B 65% C 100% D 650%
4. Last year a monthly phone bill was \$96. This year it is \$48. What is the percent decrease?
A 25% B 35% C 48% D 50%
5. Last year the price of a backpack was \$126. This year it is \$189. What is the percent increase?
A 25% B 50% C 63% D 150%
6. Last year the cost of a yearly gym membership was \$420. This year it is \$546. What is the percent increase?
A 30% B 60% C 126% D 130%
7. Last year monthly rent for an apartment was \$900. This year it is \$1,080. What is the percent increase?
A 2% B 20% C 120% D 180%
8. Last year a monthly phone bill was \$60. This year it is \$12. What is the percent decrease?
A 20% B 40% C 48% D 80%
9. Last year the price of a pair of boots was \$200. This year it is \$260. What is the percent increase?
A 15% B 30% C 60% D 130%
10. Last year the price of a pair of boots was \$60. This year it is \$30. What is the percent decrease?
A 30% B 35% C 50% D 100%
11. Last year the cost of a yearly gym membership was \$376. This year it is \$658. What is the percent increase?
A 75% B 95% C 175% D 282%
12. A store sold a camping tent for \$152. It increased the price by 25%, then put the camping tent on sale at 20% off the new price. What is the sale price?
A \$112.00 B \$121.60 C \$152.00 D \$159.60
13. A store sold a camping tent for \$296. It increased the price by 25%, then put the camping tent on sale at 10% off the new price. What is the sale price?
A \$333.00 B \$340.40 C \$370.00 D \$483.00
14. Last year the price of a pair of boots was \$152. This year it is \$228. What is the percent increase?
A 5% B 50% C 76% D 150%
15. A warehouse holds 950 crates. 60% of the crates contain medical supplies, and $\frac{1}{5}$ of those crates must be kept refrigerated. How many crates must be refrigerated?
A 104 B 114 C 190 D 570
16. Last year the price of a gallon of paint was \$72. This year it is \$90. What is the percent increase?
A 18% B 20% C 25% D 125%
17. A store sold a television for \$332. It increased the price by 25%, then put the television on sale at 20% off the new price. What is the sale price?
A \$332.00 B \$348.60 C \$415.00 D \$664.00
18. Last year monthly rent for an apartment was \$1,200. This year it is \$1,920. What is the percent increase?
A 37.5% B 60% C 160% D 720%

19. A training battalion has 400 recruits. 30% of them are assigned to the morning shift, and $\frac{1}{4}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 30 B 90 C 100 D 120
20. A pair of running shoes was priced at \$84. The store raised the price by 25%, and a month later it marked the new price down by 20%. What is the final price?
A \$67.20 B \$84.00 C \$88.20 D \$105.00
21. Last year the price of a pair of boots was \$104. This year it is \$182. What is the percent increase?
A 45% B 55% C 75% D 78%
22. A store sold a pair of running shoes for \$100. It increased the price by 40%, then put the pair of running shoes on sale at 40% off the new price. What is the sale price?
A \$60 B \$64 C \$74 D \$84
23. A store sold a bicycle for \$328. It increased the price by 25%, then put the bicycle on sale at 20% off the new price. What is the sale price?
A \$228.00 B \$262.40 C \$328.00 D \$344.40
24. A training battalion has 1,050 recruits. 40% of them are assigned to the morning shift, and $\frac{3}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 168 B 202 C 252 D 420
25. A store sold a television for \$800. It increased the price by 40%, then put the television on sale at 25% off the new price. What is the sale price?
A \$540 B \$600 C \$840 D \$920
26. A warehouse holds 950 crates. 30% of the crates contain medical supplies, and $\frac{4}{5}$ of those crates must be kept refrigerated. How many crates must be refrigerated?
A 57 B 114 C 208 D 228
27. A power drill was priced at \$228. The store raised the price by 25%, and a month later it marked the new price down by 20%. What is the final price?
A \$114.00 B \$182.40 C \$208.00 D \$228.00
28. Lance Corporal Walker surveyed 1,100 service members. 40% said they exercise every day, and $\frac{3}{5}$ of the daily exercisers said they run. How many said they run?
A 264 B 440 C 528 D 660
29. A store sold a television for \$200. It increased the price by 20%, then put the television on sale at 20% off the new price. What is the sale price?
A \$160 B \$192 C \$200 D \$240
30. A power drill was priced at \$140. The store raised the price by 20%, and a month later it marked the new price down by 25%. What is the final price?
A \$126 B \$133 C \$166 D \$168

Part 2: Mathematics Knowledge

Score: ____ / 25

Target time: 24 minutes

1. 258 is 60% of what number?
A 43 B 318 C 380 D 430
2. 60 is 75% of what number?
A 8 B 40 C 45 D 80
3. What is 25% of 284?
A 41 B 61 C 71 D 710
4. What is $\frac{16}{25}$ written as a percent?
A 6.4% B 16%
C 64% D 156.25%
5. Which of the following is equal to 0.57?
A 0.57% B 5.7% C 57% D 570%
6. What is $\frac{3}{20}$ written as a percent?
A 3% B 15% C 17% D 23%
7. 13 is what percent of 52?
A 2.5% B 25% C 30% D 400%
8. Which of the following is equal to 0.26?
A 2.6% B 26% C 46% D 260%
9. 102 is what percent of 136?
A 0.75% B 7.5% C 65% D 75%
10. What is 50% of 420?
A 21 B 105 C 170 D 210
11. 72 is what percent of 288?
A 25% B 30% C 35% D 400%
12. What percent of 380 is 570?
A 15% B 150% C 170% D 300%
13. What is 75% of 40% of 120?
A 16 B 36 C 48 D 138
14. 3 is what percent of 12?
A 0.25% B 2.5% C 5% D 25%
15. 23 is 20% of what number?
A 115 B 145 C 155 D 460
16. If 20% of a number is 91, what is the number?
A 111 B 255 C 455 D 1,820
17. What is 25% of 256?
A 64 B 192 C 320 D 640
18. If 40% of a number is 196, what is the number?
A 490 B 590 C 4,900 D 7,840
19. If 10% of a number is 2, what is the number?
A 2 B 12 C 20 D 40
20. What is 25% of 80% of 800?
A 80 B 160 C 640 D 840
21. What is 10% of 50% of 660?
A 18 B 23 C 33 D 396
22. What is 75% of 80% of 500?
A 300 B 375 C 400 D 775
23. What is 60% of 25% of 180?
A 27 B 45 C 108 D 153
24. What is 10% of 50% of 980?
A 39 B 44 C 49 D 588
25. What is 20% of 80% of 900?
A 144 B 180 C 720 D 900

Answer Key: Practice Test 1

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 6	7 B ch. 6	13 A ch. 6	19 A ch. 6	25 C ch. 6
2 A ch. 6	8 D ch. 6	14 B ch. 6	20 B ch. 6	26 D ch. 6
3 A ch. 6	9 B ch. 6	15 B ch. 6	21 C ch. 6	27 D ch. 6
4 D ch. 6	10 C ch. 6	16 C ch. 6	22 D ch. 6	28 A ch. 6
5 B ch. 6	11 A ch. 6	17 A ch. 6	23 C ch. 6	29 B ch. 6
6 A ch. 6	12 C ch. 6	18 B ch. 6	24 C ch. 6	30 A ch. 6

Part 2: Mathematics Knowledge

1 D ch. 6	6 B ch. 6	11 A ch. 6	16 C ch. 6	21 C ch. 6
2 D ch. 6	7 B ch. 6	12 B ch. 6	17 A ch. 6	22 A ch. 6
3 C ch. 6	8 B ch. 6	13 B ch. 6	18 A ch. 6	23 A ch. 6
4 C ch. 6	9 D ch. 6	14 D ch. 6	19 C ch. 6	24 C ch. 6
5 C ch. 6	10 D ch. 6	15 A ch. 6	20 B ch. 6	25 A ch. 6

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** 40%

- Find the amount of decrease: $1,900 - 1,140 = 760$ dollars.
- Divide by the *original* price: $\frac{760}{1,900} = 0.4$.
- Convert to a percent: $0.4 = 40\%$ decrease.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

2. **A** 25%

- Find the amount of decrease: $1,500 - 1,125 = 375$ dollars.
- Divide by the *original* price: $\frac{375}{1,500} = 0.25$.
- Convert to a percent: $0.25 = 25\%$ decrease.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

3. **A** 50%
- Find the amount of decrease: $1,300 - 650 = 650$ dollars.
 - Divide by the *original* price: $\frac{650}{1,300} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ decrease.
- WHY NOT** (C) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.
4. **D** 50%
- Find the amount of decrease: $96 - 48 = 48$ dollars.
 - Divide by the *original* price: $\frac{48}{96} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ decrease.
- WHY NOT** (C) gives the dollar change instead of the percent change.
5. **B** 50%
- Find the amount of increase: $189 - 126 = 63$ dollars.
 - Divide by the *original* price: $\frac{63}{126} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ increase.
- WHY NOT** (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.
6. **A** 30%
- Find the amount of increase: $546 - 420 = 126$ dollars.
 - Divide by the *original* price: $\frac{126}{420} = 0.3$.
 - Convert to a percent: $0.3 = 30\%$ increase.
- WHY NOT** (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.
7. **B** 20%
- Find the amount of increase: $1,080 - 900 = 180$ dollars.
 - Divide by the *original* price: $\frac{180}{900} = 0.2$.
 - Convert to a percent: $0.2 = 20\%$ increase.
- WHY NOT** (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.
8. **D** 80%
- Find the amount of decrease: $60 - 12 = 48$ dollars.
 - Divide by the *original* price: $\frac{48}{60} = 0.8$.
 - Convert to a percent: $0.8 = 80\%$ decrease.
- WHY NOT** (A) gives the new price as a percent of the old price. (C) gives the dollar change instead of the percent change.
9. **B** 30%
- Find the amount of increase: $260 - 200 = 60$ dollars.
 - Divide by the *original* price: $\frac{60}{200} = 0.3$.
 - Convert to a percent: $0.3 = 30\%$ increase.
- WHY NOT** (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.
10. **C** 50%
- Find the amount of decrease: $60 - 30 = 30$ dollars.
 - Divide by the *original* price: $\frac{30}{60} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ decrease.
- WHY NOT** (A) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.
11. **A** 75%
- Find the amount of increase: $658 - 376 = 282$ dollars.
 - Divide by the *original* price: $\frac{282}{376} = 0.75$.
 - Convert to a percent: $0.75 = 75\%$ increase.
- WHY NOT** (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.
12. **C** \$152.00
- Raise the price by 25%: $152 \times 1.25 = 190$.
 - Take 20% off the *new* price: $190 \times 0.8 = 152$.
 - The final price is \$152.
- WHY NOT** (B) applies only the decrease. (D) combines the changes into a single 5% increase.
13. **A** \$333.00
- Raise the price by 25%: $296 \times 1.25 = 370$.
 - Take 10% off the *new* price: $370 \times 0.9 = 333$.
 - The final price is \$333.
- WHY NOT** (B) combines the changes into a single 15% increase. (C) stops after the increase.
14. **B** 50%
- Find the amount of increase: $228 - 152 = 76$ dollars.
 - Divide by the *original* price: $\frac{76}{152} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ increase.
- WHY NOT** (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

15. **B** 114

- 1 Find the first group: $0.6 \times 950 = 570$.
- 2 Take $\frac{1}{5}$ of that group: $\frac{1}{5} \times 570 = 114$.

WHY NOT (C) takes $\frac{1}{5}$ of the whole total. (D) is the size of the first group only.

16. **C** 25%

- 1 Find the amount of increase: $90 - 72 = 18$ dollars.
- 2 Divide by the *original* price: $\frac{18}{72} = 0.25$.
- 3 Convert to a percent: $0.25 = 25\%$ increase.

WHY NOT (A) gives the dollar change instead of the percent change. (B) divides by the new price instead of the original price. (D) gives the new price as a percent of the old price.

17. **A** \$332.00

- 1 Raise the price by 25%: $332 \times 1.25 = 415$.
- 2 Take 20% off the *new* price: $415 \times 0.8 = 332$.
- 3 The final price is \$332.

WHY NOT (B) combines the changes into a single 5% increase. (C) stops after the increase.

18. **B** 60%

- 1 Find the amount of increase: $1,920 - 1,200 = 720$ dollars.
- 2 Divide by the *original* price: $\frac{720}{1,200} = 0.6$.
- 3 Convert to a percent: $0.6 = 60\%$ increase.

WHY NOT (A) divides by the new price instead of the original price. (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

19. **A** 30

- 1 Find the first group: $0.3 \times 400 = 120$.
- 2 Take $\frac{1}{4}$ of that group: $\frac{1}{4} \times 120 = 30$.

WHY NOT (B) is the part of the group that is *not* included. (C) takes $\frac{1}{4}$ of the whole total. (D) is the size of the first group only.

20. **B** \$84.00

- 1 Raise the price by 25%: $84 \times 1.25 = 105$.
- 2 Take 20% off the *new* price: $105 \times 0.8 = 84$.
- 3 The final price is \$84.

WHY NOT (A) applies only the decrease. (C) combines the changes into a single 5% increase. (D) stops after the increase.

21. **C** 75%

- 1 Find the amount of increase: $182 - 104 = 78$ dollars.
- 2 Divide by the *original* price: $\frac{78}{104} = 0.75$.
- 3 Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

22. **D** \$84

- 1 Raise the price by 40%: $100 \times 1.4 = 140$.
- 2 Take 40% off the *new* price: $140 \times 0.6 = 84$.
- 3 The final price is \$84.

WHY NOT (A) applies only the decrease.

23. **C** \$328.00

- 1 Raise the price by 25%: $328 \times 1.25 = 410$.
- 2 Take 20% off the *new* price: $410 \times 0.8 = 328$.
- 3 The final price is \$328.

WHY NOT (B) applies only the decrease. (D) combines the changes into a single 5% increase.

24. **C** 252

- 1 Find the first group: $0.4 \times 1,050 = 420$.
- 2 Take $\frac{3}{5}$ of that group: $\frac{3}{5} \times 420 = 252$.

WHY NOT (A) is the part of the group that is *not* included. (D) is the size of the first group only.

25. **C** \$840

- 1 Raise the price by 40%: $800 \times 1.4 = 1,120$.
- 2 Take 25% off the *new* price: $1,120 \times 0.75 = 840$.
- 3 The final price is \$840.

WHY NOT (B) applies only the decrease. (D) combines the changes into a single 15% increase.

26. **D** 228

- 1 Find the first group: $0.3 \times 950 = 285$.
- 2 Take $\frac{4}{5}$ of that group: $\frac{4}{5} \times 285 = 228$.

WHY NOT (A) is the part of the group that is *not* included.

27. **D** \$228.00

- 1 Raise the price by 25%: $228 \times 1.25 = 285$.
- 2 Take 20% off the *new* price: $285 \times 0.8 = 228$.
- 3 The final price is \$228.

WHY NOT (B) applies only the decrease.

28. **A** 264

- 1 Find the first group: $0.4 \times 1,100 = 440$.
- 2 Take $\frac{3}{5}$ of that group: $\frac{3}{5} \times 440 = 264$.

WHY NOT (B) is the size of the first group only. (D) takes $\frac{3}{5}$ of the whole total.

29. **B** \$192

- 1 Raise the price by 20%: $200 \times 1.2 = 240$.
- 2 Take 20% off the *new* price: $240 \times 0.8 = 192$.
- 3 The final price is \$192.

WHY NOT (A) applies only the decrease. (C) assumes the two changes cancel out. (D) stops after the increase.

30. **A** \$126

- 1 Raise the price by 20%: $140 \times 1.2 = 168$.
- 2 Take 25% off the *new* price: $168 \times 0.75 = 126$.
- 3 The final price is \$126.

WHY NOT (B) combines the changes into a single 5% decrease. (D) stops after the increase.

Explanations: Mathematics Knowledge

1. **D** 430

- 1 Use part = percent $\times$ whole: $258 = 0.6 \times w$.
- 2 Divide both sides by 0.6: $w = 258 \div 0.6 = 430$.
- 3 Check: 60% of 430 is 258. ✓

WHY NOT (A) slips the decimal point. (B) adds the percent to the part.

2. **D** 80

- 1 Use part = percent $\times$ whole: $60 = 0.75 \times w$.
- 2 Divide both sides by 0.75: $w = 60 \div 0.75 = 80$.
- 3 Check: 75% of 80 is 60. ✓

WHY NOT (A) slips the decimal point. (C) finds 75% of 60 instead.

3. **C** 71

- 1 Change the percent to a decimal: $25\% = 0.25$.
- 2 Multiply by the number: $0.25 \times 284 = 71$.

WHY NOT (D) moves the decimal point only one place.

4. **C** 64%

- 1 Divide the numerator by the denominator: $16 \div 25 = 0.64$.
- 2 Multiply by 100 to get a percent: $0.64 = 64\%$.

TIP Or scale the fraction to hundredths: $\frac{16}{25} = \frac{64}{100}$.

WHY NOT (A) moves the decimal point only one place. (B) just writes the numerator as if it were a percent. (D) divides the denominator by the numerator.

5. **C** 57%

- 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- 2 $0.57 \times 100 = 57$, so $0.57 = 57\%$.

WHY NOT (A) forgets to move the decimal point at all. (B) moves the decimal point only one place. (D) moves the decimal point three places.

6. **B** 15%

- 1 Divide the numerator by the denominator: $3 \div 20 = 0.15$.
- 2 Multiply by 100 to get a percent: $0.15 = 15\%$.

TIP Or scale the fraction to hundredths: $\frac{3}{20} = \frac{15}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent.

7. **B** 25%

- 1 Write the part over the whole: $\frac{13}{52}$.
- 2 Simplify and convert: $\frac{13}{52} = 0.25 = 25\%$.

WHY NOT (A) moves the decimal point only one place. (D) divides the whole by the part instead of the part by the whole.

8. **B** 26%

- 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- 2 $0.26 \times 100 = 26$, so $0.26 = 26\%$.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.

9. **D** 75%

- 1 Write the part over the whole: $\frac{102}{136}$.
- 2 Simplify and convert: $\frac{102}{136} = 0.75 = 75\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

10. **D** 210

- 1 Change the percent to a decimal: $50\% = 0.5$.
- 2 Multiply by the number: $0.5 \times 420 = 210$.

WHY NOT (A) moves the decimal point three places.

11. **A** 25%

- 1 Write the part over the whole: $\frac{72}{288}$.
- 2 Simplify and convert: $\frac{72}{288} = 0.25 = 25\%$.

WHY NOT (D) divides the whole by the part instead of the part by the whole.

12. **B** 150%

- 1 Write the part over the whole: $\frac{570}{380}$.
- 2 Simplify and convert: $\frac{570}{380} = 1.5 = 150\%$.

WHY NOT (A) moves the decimal point only one place.

13. **B** 36

- 1 Work from the inside out. First, 40% of 120: $0.4 \times 120 = 48$.
- 2 Then 75% of that result: $0.75 \times 48 = 36$.

TIP Percents of percents multiply: $0.75 \times 0.4 = 0.3$, and $0.3 \times 120 = 36$.

WHY NOT (C) stops after taking 40%. (D) adds the two percents.

14. **D** 25%

- 1 Write the part over the whole: $\frac{3}{12}$.
- 2 Simplify and convert: $\frac{3}{12} = 0.25 = 25\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

15. **A** 115

- 1 Use part = percent $\times$ whole: $23 = 0.2 \times w$.
- 2 Divide both sides by 0.2: $w = 23 \div 0.2 = 115$.
- 3 Check: 20% of 115 is 23. ✓

WHY NOT (D) multiplies by the percent without converting it.

16. **C** 455

- 1 Use part = percent $\times$ whole: $91 = 0.2 \times w$.
- 2 Divide both sides by 0.2: $w = 91 \div 0.2 = 455$.
- 3 Check: 20% of 455 is 91. ✓

WHY NOT (A) adds the percent to the part. (D) multiplies by the percent without converting it.

17. **A** 64

- 1 Change the percent to a decimal: $25\% = 0.25$.
- 2 Multiply by the number: $0.25 \times 256 = 64$.

WHY NOT (B) is what is left after removing 25%, not 25% of the number. (C) adds 25% to the number. (D) moves the decimal point only one place.

18. **A** 490

- 1 Use part = percent $\times$ whole: $196 = 0.4 \times w$.
- 2 Divide both sides by 0.4: $w = 196 \div 0.4 = 490$.
- 3 Check: 40% of 490 is 196. ✓

WHY NOT (D) multiplies by the percent without converting it.

19. **C** 20

- 1 Use part = percent $\times$ whole: $2 = 0.1 \times w$.
- 2 Divide both sides by 0.1: $w = 2 \div 0.1 = 20$.
- 3 Check: 10% of 20 is 2. ✓

WHY NOT (A) slips the decimal point. (B) adds the percent to the part.

20. **B** 160

- 1 Work from the inside out. First, 80% of 800: $0.8 \times 800 = 640$.
- 2 Then 25% of that result: $0.25 \times 640 = 160$.

TIP Percents of percents multiply: $0.25 \times 0.8 = 0.2$, and $0.2 \times 800 = 160$.

WHY NOT (C) stops after taking 80%. (D) adds the two percents.

21. **C** 33

- 1 Work from the inside out. First, 50% of 660: $0.5 \times 660 = 330$.
- 2 Then 10% of that result: $0.1 \times 330 = 33$.

TIP Percents of percents multiply: $0.1 \times 0.5 = 0.05$, and $0.05 \times 660 = 33$.

WHY NOT (D) adds the two percents.

22. **A** 300

- 1 Work from the inside out. First, 80% of 500: $0.8 \times 500 = 400$.
- 2 Then 75% of that result: $0.75 \times 400 = 300$.

TIP Percents of percents multiply: $0.75 \times 0.8 = 0.6$, and $0.6 \times 500 = 300$.

WHY NOT (B) takes only 75% of 500. (C) stops after taking 80%. (D) adds the two percents.

23. **A** 27

- 1 Work from the inside out. First, 25% of 180: $0.25 \times 180 = 45$.
- 2 Then 60% of that result: $0.6 \times 45 = 27$.

TIP Percents of percents multiply: $0.6 \times 0.25 = 0.15$, and $0.15 \times 180 = 27$.

WHY NOT (B) stops after taking 25%. (C) takes only 60% of 180. (D) adds the two percents.

24. **C** 49

- 1 Work from the inside out. First, 50% of 980: $0.5 \times 980 = 490$.
- 2 Then 10% of that result: $0.1 \times 490 = 49$.

TIP Percents of percents multiply: $0.1 \times 0.5 = 0.05$, and $0.05 \times 980 = 49$.

WHY NOT (D) adds the two percents.

25. **A** 144

1 Work from the inside out. First, 80% of 900:
 $0.8 \times 900 = 720$.

2 Then 20% of that result: $0.2 \times 720 = 144$.

TIP Percents of percents multiply: $0.2 \times 0.8 = 0.16$,
and $0.16 \times 900 = 144$.

WHY NOT (B) takes only 20% of 900. (C) stops after
taking 80%. (D) adds the two percents.

Practice Test 2

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
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- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
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- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

1. Last year the price of a pair of boots was \$148. This year it is \$74. What is the percent decrease?
A 30% B 50% C 74% D 100%
2. Last year the price of a used car was \$8,900. This year it is \$4,450. What is the percent decrease?
A 5% B 50% C 100% D 4,450%
3. Last year the price of a tool kit was \$112. This year it is \$140. What is the percent increase?
A 25% B 28% C 125% D 250%
4. Last year the price of a used car was \$8,300. This year it is \$11,620. What is the percent increase?
A 40% B 80% C 140% D 3,320%
5. Last year monthly rent for an apartment was \$1,300. This year it is \$1,690. What is the percent increase?
A 30% B 130% C 300% D 390%
6. Last year monthly rent for an apartment was \$1,000. This year it is \$500. What is the percent decrease?
A 50% B 55% C 100% D 500%
7. Last year the price of a pair of boots was \$128. This year it is \$64. What is the percent decrease?
A 50% B 64% C 65% D 100%
8. Last year the price of a pair of boots was \$192. This year it is \$144. What is the percent decrease?
A 5% B 25% C 48% D 75%
9. Last year the price of a tool kit was \$220. This year it is \$176. What is the percent decrease?
A 20% B 25% C 44% D 80%
10. Last year monthly rent for an apartment was \$1,000. This year it is \$1,400. What is the percent increase?
A 4% B 40% C 140% D 400%
11. Last year a monthly phone bill was \$80. This year it is \$120. What is the percent increase?
A 50% B 60% C 70% D 150%
12. Last year monthly rent for an apartment was \$900. This year it is \$360. What is the percent decrease?
A 6% B 30% C 40% D 60%
13. Last year the price of a pair of boots was \$160. This year it is \$224. What is the percent increase?
A 40% B 50% C 64% D 140%
14. A store sold a bicycle for \$240. It increased the price by 25%, then put the bicycle on sale at 20% off the new price. What is the sale price?
A \$180 B \$192 C \$240 D \$252
15. Last year monthly rent for an apartment was \$1,500. This year it is \$2,625. What is the percent increase?
A 35% B 65% C 75% D 1,125%
16. Last year the price of a pair of boots was \$160. This year it is \$208. What is the percent increase?
A 3% B 25% C 30% D 48%
17. Last year the price of a used car was \$10,700. This year it is \$16,050. What is the percent increase?
A 5% B 50% C 150% D 5,350%
18. Last year the price of a used car was \$7,100. This year it is \$9,230. What is the percent increase?
A 3% B 15% C 30% D 2,130%

19. A warehouse holds 300 crates. 80% of the crates contain medical supplies, and $\frac{1}{3}$ of those crates must be kept refrigerated. How many crates must be refrigerated?
A 8 B 80 C 100 D 240
20. Last year a monthly phone bill was \$60. This year it is \$66. What is the percent increase?
A 10% B 11% C 20% D 110%
21. Last year the price of a gallon of paint was \$20. This year it is \$35. What is the percent increase?
A 15% B 65% C 75% D 175%
22. A store sold a bicycle for \$400. It increased the price by 50%, then put the bicycle on sale at 25% off the new price. What is the sale price?
A \$225 B \$250 C \$300 D \$450
23. A store sold a bicycle for \$496. It increased the price by 25%, then put the bicycle on sale at 40% off the new price. What is the sale price?
A \$272.00 B \$297.60 C \$322.00 D \$372.00
24. A warehouse holds 600 crates. 20% of the crates contain medical supplies, and $\frac{1}{3}$ of those crates must be kept refrigerated. How many crates must be refrigerated?
A 20 B 25 C 40 D 120
25. Last year the price of a backpack was \$150. This year it is \$210. What is the percent increase?
A 20% B 40% C 60% D 140%
26. A warehouse holds 100 crates. 45% of the crates contain medical supplies, and $\frac{4}{5}$ of those crates must be kept refrigerated. How many crates must be refrigerated?
A 9 B 21 C 36 D 45
27. Corporal Nguyen surveyed 900 service members. 40% said they exercise every day, and $\frac{1}{4}$ of the daily exercisers said they run. How many said they run?
A 90 B 225 C 270 D 360
28. A store sold a bicycle for \$568. It increased the price by 25%, then put the bicycle on sale at 50% off the new price. What is the sale price?
A \$155 B \$284 C \$305 D \$355
29. Seaman Kim surveyed 450 service members. 60% said they exercise every day, and $\frac{1}{2}$ of the daily exercisers said they run. How many said they run?
A 55 B 135 C 225 D 270
30. Petty Officer Brooks surveyed 800 service members. 30% said they exercise every day, and $\frac{1}{5}$ of the daily exercisers said they run. How many said they run?
A 38 B 48 C 160 D 240

Part 2: Mathematics Knowledge

Score: ____ / 25

Target time: 24 minutes

1. Which of the following is equal to 0.06?
A 0.6% B 6% C 7.5% D 60%
2. What is $\frac{5}{8}$ written as a percent?
A 62.5% B 63% C 125% D 160%
3. What percent of 384 is 288?
A 0.75% B 7.5% C 75% D 95%
4. What percent of 240 is 192?
A 0.8% B 8% C 80% D 125%
5. If 75% of a number is 315, what is the number?
A 420 B 620 C 4,200 D 23,625
6. What percent of 380 is 57?
A 0.15% B 1.5% C 11% D 15%
7. What is $\frac{1}{20}$ written as a percent?
A 5% B 6% C 21% D 2,000%
8. What is $\frac{2}{25}$ written as a percent?
A 0.8% B 2% C 8% D 1,250%
9. 48 is what percent of 160?
A 0.3% B 3% C 25% D 30%
10. What is $\frac{2}{5}$ written as a percent?
A 2% B 4% C 7% D 40%
11. What is 150% of 888?
A 1,332 B 2,220 C 2,664 D 13,320
12. If 25% of a number is 60, what is the number?
A 15 B 24 C 85 D 240
13. What is $\frac{4}{5}$ written as a percent?
A 4% B 80% C 100% D 125%
14. If 40% of a number is 32, what is the number?
A 8 B 72 C 80 D 1,280
15. What is 25% of 20% of 220?
A 7 B 10 C 11 D 99
16. What is 75% of 80% of 520?
A 26 B 312 C 416 D 806
17. What is 150% of 752?
A 728 B 828 C 1,128 D 11,280
18. What is 80% of 300?
A 240 B 280 C 540 D 2,400
19. What is 60% of 75% of 280?
A 42 B 126 C 210 D 378
20. What is $\frac{1}{5}$ written as a percent?
A 1% B 2% C 6% D 20%
21. What is 50% of 80% of 880?
A 264 B 352 C 704 D 1,144
22. What is 75% of 80% of 860?
A 43 B 366 C 516 D 1,333
23. What is 50% of 60% of 880?
A 88 B 264 C 528 D 968
24. What is 20% of 75% of 540?
A 81 B 108 C 405 D 513
25. What is 50% of 75% of 200?
A 75 B 100 C 150 D 250

Answer Key: Practice Test 2

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 6	7 A ch. 6	13 A ch. 6	19 B ch. 6	25 B ch. 6
2 B ch. 6	8 B ch. 6	14 C ch. 6	20 A ch. 6	26 C ch. 6
3 A ch. 6	9 A ch. 6	15 C ch. 6	21 C ch. 6	27 A ch. 6
4 A ch. 6	10 B ch. 6	16 C ch. 6	22 D ch. 6	28 D ch. 6
5 A ch. 6	11 A ch. 6	17 B ch. 6	23 D ch. 6	29 B ch. 6
6 A ch. 6	12 D ch. 6	18 C ch. 6	24 C ch. 6	30 B ch. 6

Part 2: Mathematics Knowledge

1 B ch. 6	6 D ch. 6	11 A ch. 6	16 B ch. 6	21 B ch. 6
2 A ch. 6	7 A ch. 6	12 D ch. 6	17 C ch. 6	22 C ch. 6
3 C ch. 6	8 C ch. 6	13 B ch. 6	18 A ch. 6	23 B ch. 6
4 C ch. 6	9 D ch. 6	14 C ch. 6	19 B ch. 6	24 A ch. 6
5 A ch. 6	10 D ch. 6	15 C ch. 6	20 D ch. 6	25 A ch. 6

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** 50%

- Find the amount of decrease: $148 - 74 = 74$ dollars.
- Divide by the *original* price: $\frac{74}{148} = 0.5$.
- Convert to a percent: $0.5 = 50\%$ decrease.

WHY NOT (C) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.

2. **B** 50%

- Find the amount of decrease: $8,900 - 4,450 = 4,450$ dollars.
- Divide by the *original* price: $\frac{4,450}{8,900} = 0.5$.
- Convert to a percent: $0.5 = 50\%$ decrease.

WHY NOT (C) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.

3. **A** 25%
- Find the amount of increase: $140 - 112 = 28$ dollars.
 - Divide by the *original* price: $\frac{28}{112} = 0.25$.
 - Convert to a percent: $0.25 = 25\%$ increase.
- WHY NOT** (B) gives the dollar change instead of the percent change. (C) gives the new price as a percent of the old price.
4. **A** 40%
- Find the amount of increase: $11,620 - 8,300 = 3,320$ dollars.
 - Divide by the *original* price: $\frac{3,320}{8,300} = 0.4$.
 - Convert to a percent: $0.4 = 40\%$ increase.
- WHY NOT** (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.
5. **A** 30%
- Find the amount of increase: $1,690 - 1,300 = 390$ dollars.
 - Divide by the *original* price: $\frac{390}{1,300} = 0.3$.
 - Convert to a percent: $0.3 = 30\%$ increase.
- WHY NOT** (B) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.
6. **A** 50%
- Find the amount of decrease: $1,000 - 500 = 500$ dollars.
 - Divide by the *original* price: $\frac{500}{1,000} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ decrease.
- WHY NOT** (C) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.
7. **A** 50%
- Find the amount of decrease: $128 - 64 = 64$ dollars.
 - Divide by the *original* price: $\frac{64}{128} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ decrease.
- WHY NOT** (B) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.
8. **B** 25%
- Find the amount of decrease: $192 - 144 = 48$ dollars.
 - Divide by the *original* price: $\frac{48}{192} = 0.25$.
 - Convert to a percent: $0.25 = 25\%$ decrease.
- WHY NOT** (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.
9. **A** 20%
- Find the amount of decrease: $220 - 176 = 44$ dollars.
 - Divide by the *original* price: $\frac{44}{220} = 0.2$.
 - Convert to a percent: $0.2 = 20\%$ decrease.
- WHY NOT** (B) divides by the new price instead of the original price. (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.
10. **B** 40%
- Find the amount of increase: $1,400 - 1,000 = 400$ dollars.
 - Divide by the *original* price: $\frac{400}{1,000} = 0.4$.
 - Convert to a percent: $0.4 = 40\%$ increase.
- WHY NOT** (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.
11. **A** 50%
- Find the amount of increase: $120 - 80 = 40$ dollars.
 - Divide by the *original* price: $\frac{40}{80} = 0.5$.
 - Convert to a percent: $0.5 = 50\%$ increase.
- WHY NOT** (D) gives the new price as a percent of the old price.
12. **D** 60%
- Find the amount of decrease: $900 - 360 = 540$ dollars.
 - Divide by the *original* price: $\frac{540}{900} = 0.6$.
 - Convert to a percent: $0.6 = 60\%$ decrease.
- WHY NOT** (C) gives the new price as a percent of the old price.
13. **A** 40%
- Find the amount of increase: $224 - 160 = 64$ dollars.
 - Divide by the *original* price: $\frac{64}{160} = 0.4$.
 - Convert to a percent: $0.4 = 40\%$ increase.
- WHY NOT** (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.
14. **C** \$240
- Raise the price by 25%: $240 \times 1.25 = 300$.
 - Take 20% off the *new* price: $300 \times 0.8 = 240$.
 - The final price is \$240.
- WHY NOT** (B) applies only the decrease. (D) combines the changes into a single 5% increase.

15. **C** 75%

- Find the amount of increase: $2,625 - 1,500 = 1,125$ dollars.
- Divide by the *original* price: $\frac{1,125}{1,500} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

16. **C** 30%

- Find the amount of increase: $208 - 160 = 48$ dollars.
- Divide by the *original* price: $\frac{48}{160} = 0.3$.
- Convert to a percent: $0.3 = 30\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

17. **B** 50%

- Find the amount of increase: $16,050 - 10,700 = 5,350$ dollars.
- Divide by the *original* price: $\frac{5,350}{10,700} = 0.5$.
- Convert to a percent: $0.5 = 50\%$ increase.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

18. **C** 30%

- Find the amount of increase: $9,230 - 7,100 = 2,130$ dollars.
- Divide by the *original* price: $\frac{2,130}{7,100} = 0.3$.
- Convert to a percent: $0.3 = 30\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

19. **B** 80

- Find the first group: $0.8 \times 300 = 240$.
- Take $\frac{1}{3}$ of that group: $\frac{1}{3} \times 240 = 80$.

WHY NOT (C) takes $\frac{1}{3}$ of the whole total. (D) is the size of the first group only.

20. **A** 10%

- Find the amount of increase: $66 - 60 = 6$ dollars.
- Divide by the *original* price: $\frac{6}{60} = 0.1$.
- Convert to a percent: $0.1 = 10\%$ increase.

WHY NOT (D) gives the new price as a percent of the old price.

21. **C** 75%

- Find the amount of increase: $35 - 20 = 15$ dollars.
- Divide by the *original* price: $\frac{15}{20} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (A) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

22. **D** \$450

- Raise the price by 50%: $400 \times 1.5 = 600$.
- Take 25% off the *new* price: $600 \times 0.75 = 450$.
- The final price is \$450.

WHY NOT (C) applies only the decrease.

23. **D** \$372.00

- Raise the price by 25%: $496 \times 1.25 = 620$.
- Take 40% off the *new* price: $620 \times 0.6 = 372$.
- The final price is \$372.

WHY NOT (B) applies only the decrease.

24. **C** 40

- Find the first group: $0.2 \times 600 = 120$.
- Take $\frac{1}{3}$ of that group: $\frac{1}{3} \times 120 = 40$.

WHY NOT (D) is the size of the first group only.

25. **B** 40%

- Find the amount of increase: $210 - 150 = 60$ dollars.
- Divide by the *original* price: $\frac{60}{150} = 0.4$.
- Convert to a percent: $0.4 = 40\%$ increase.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

26. **C** 36

- Find the first group: $0.45 \times 100 = 45$.
- Take $\frac{4}{5}$ of that group: $\frac{4}{5} \times 45 = 36$.

WHY NOT (A) is the part of the group that is *not* included. (D) is the size of the first group only.

27. **A** 90

- Find the first group: $0.4 \times 900 = 360$.
- Take $\frac{1}{4}$ of that group: $\frac{1}{4} \times 360 = 90$.

WHY NOT (B) takes $\frac{1}{4}$ of the whole total. (C) is the part of the group that is *not* included. (D) is the size of the first group only.

28. **D** \$355

- Raise the price by 25%: $568 \times 1.25 = 710$.
- Take 50% off the *new* price: $710 \times 0.5 = 355$.
- The final price is \$355.

WHY NOT (B) applies only the decrease.

29. **B** 135

- Find the first group: $0.6 \times 450 = 270$.
- Take $\frac{1}{2}$ of that group: $\frac{1}{2} \times 270 = 135$.

WHY NOT (C) takes $\frac{1}{2}$ of the whole total. (D) is the size of the first group only.

30. **B** 48

- Find the first group: $0.3 \times 800 = 240$.
- Take $\frac{1}{5}$ of that group: $\frac{1}{5} \times 240 = 48$.

WHY NOT (C) takes $\frac{1}{5}$ of the whole total. (D) is the size of the first group only.

Explanations: Mathematics Knowledge

1. **B** 6%

- To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- $0.06 \times 100 = 6$, so $0.06 = 6\%$.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.

2. **A** 62.5%

- Divide the numerator by the denominator: $5 \div 8 = 0.625$.
- Multiply by 100 to get a percent: $0.625 = 62.5\%$.

WHY NOT (D) divides the denominator by the numerator.

3. **C** 75%

- Write the part over the whole: $\frac{288}{384}$.
- Simplify and convert: $\frac{288}{384} = 0.75 = 75\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

4. **C** 80%

- Write the part over the whole: $\frac{192}{240}$.
- Simplify and convert: $\frac{192}{240} = 0.8 = 80\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place. (D) divides the whole by the part instead of the part by the whole.

5. **A** 420

- Use part = percent $\times$ whole: $315 = 0.75 \times w$.
- Divide both sides by 0.75: $w = 315 \div 0.75 = 420$.
- Check: 75% of 420 is 315. $\checkmark$

WHY NOT (D) multiplies by the percent without converting it.

6. **D** 15%

- Write the part over the whole: $\frac{57}{380}$.
- Simplify and convert: $\frac{57}{380} = 0.15 = 15\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

7. **A** 5%

- Divide the numerator by the denominator: $1 \div 20 = 0.05$.

- Multiply by 100 to get a percent: $0.05 = 5\%$.

TIP Or scale the fraction to hundredths: $\frac{1}{20} = \frac{5}{100}$.

WHY NOT (D) divides the denominator by the numerator.

8. **C** 8%

- Divide the numerator by the denominator: $2 \div 25 = 0.08$.

- Multiply by 100 to get a percent: $0.08 = 8\%$.

TIP Or scale the fraction to hundredths: $\frac{2}{25} = \frac{8}{100}$.

WHY NOT (A) moves the decimal point only one place. (B) just writes the numerator as if it were a percent. (D) divides the denominator by the numerator.

9. **D** 30%

- Write the part over the whole: $\frac{48}{160}$.

- Simplify and convert: $\frac{48}{160} = 0.3 = 30\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

10. **D** 40%

- Divide the numerator by the denominator: $2 \div 5 = 0.4$.

- Multiply by 100 to get a percent: $0.4 = 40\%$.

TIP Or scale the fraction to hundredths: $\frac{2}{5} = \frac{40}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place.

11. **A** 1,332

- Change the percent to a decimal: $150\% = 1.5$.
- Multiply by the number: $1.5 \times 888 = 1,332$.

WHY NOT (B) adds 150% to the number. (D) moves the decimal point only one place.

12. **D** 240

- Use part = percent $\times$ whole: $60 = 0.25 \times w$.
- Divide both sides by 0.25: $w = 60 \div 0.25 = 240$.
- Check: 25% of 240 is 60. $\checkmark$

WHY NOT (A) finds 25% of 60 instead. (B) slips the decimal point. (C) adds the percent to the part.

13. **B** 80%

1 Divide the numerator by the denominator: $4 \div 5 = 0.8$.

2 Multiply by 100 to get a percent: $0.8 = 80\%$.

TIP Or scale the fraction to hundredths: $\frac{4}{5} = \frac{80}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent. (D) divides the denominator by the numerator.

14. **C** 80

1 Use part = percent $\times$ whole: $32 = 0.4 \times w$.

2 Divide both sides by 0.4: $w = 32 \div 0.4 = 80$.

3 Check: 40% of 80 is 32. ✓

WHY NOT (A) slips the decimal point. (B) adds the percent to the part. (D) multiplies by the percent without converting it.

15. **C** 11

1 Work from the inside out. First, 20% of 220: $0.2 \times 220 = 44$.

2 Then 25% of that result: $0.25 \times 44 = 11$.

TIP Percents of percents multiply: $0.25 \times 0.2 = 0.05$, and $0.05 \times 220 = 11$.

WHY NOT (D) adds the two percents.

16. **B** 312

1 Work from the inside out. First, 80% of 520: $0.8 \times 520 = 416$.

2 Then 75% of that result: $0.75 \times 416 = 312$.

TIP Percents of percents multiply: $0.75 \times 0.8 = 0.6$, and $0.6 \times 520 = 312$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 80%. (D) adds the two percents.

17. **C** 1,128

1 Change the percent to a decimal: $150\% = 1.5$.

2 Multiply by the number: $1.5 \times 752 = 1,128$.

WHY NOT (D) moves the decimal point only one place.

18. **A** 240

1 Change the percent to a decimal: $80\% = 0.8$.

2 Multiply by the number: $0.8 \times 300 = 240$.

WHY NOT (C) adds 80% to the number. (D) moves the decimal point only one place.

19. **B** 126

1 Work from the inside out. First, 75% of 280: $0.75 \times 280 = 210$.

2 Then 60% of that result: $0.6 \times 210 = 126$.

TIP Percents of percents multiply: $0.6 \times 0.75 = 0.45$, and $0.45 \times 280 = 126$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 75%. (D) adds the two percents.

20. **D** 20%

1 Divide the numerator by the denominator: $1 \div 5 = 0.2$.

2 Multiply by 100 to get a percent: $0.2 = 20\%$.

TIP Or scale the fraction to hundredths: $\frac{1}{5} = \frac{20}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place.

21. **B** 352

1 Work from the inside out. First, 80% of 880: $0.8 \times 880 = 704$.

2 Then 50% of that result: $0.5 \times 704 = 352$.

TIP Percents of percents multiply: $0.5 \times 0.8 = 0.4$, and $0.4 \times 880 = 352$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 80%. (D) adds the two percents.

22. **C** 516

1 Work from the inside out. First, 80% of 860: $0.8 \times 860 = 688$.

2 Then 75% of that result: $0.75 \times 688 = 516$.

TIP Percents of percents multiply: $0.75 \times 0.8 = 0.6$, and $0.6 \times 860 = 516$.

WHY NOT (A) subtracts the two percents. (D) adds the two percents.

23. **B** 264

1 Work from the inside out. First, 60% of 880: $0.6 \times 880 = 528$.

2 Then 50% of that result: $0.5 \times 528 = 264$.

TIP Percents of percents multiply: $0.5 \times 0.6 = 0.3$, and $0.3 \times 880 = 264$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 60%. (D) adds the two percents.

24. **A** 81

1 Work from the inside out. First, 75% of 540: $0.75 \times 540 = 405$.

2 Then 20% of that result: $0.2 \times 405 = 81$.

TIP Percents of percents multiply: $0.2 \times 0.75 = 0.15$, and $0.15 \times 540 = 81$.

WHY NOT (B) takes only 20% of 540. (C) stops after taking 75%. (D) adds the two percents.

25. **A** 75

1 Work from the inside out. First, 75% of 200: $0.75 \times 200 = 150$.

2 Then 50% of that result: $0.5 \times 150 = 75$.

WHY NOT (B) takes only 50% of 200. (C) stops after taking 75%. (D) adds the two percents.

Practice Test 3

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1

A

B

C

D
- 2

A

B

C

D
- 3

A

B

C

D
- 4

A

B

C

D
- 5

A

B

C

D
- 6

A

B

C

D
- 7

A

B

C

D
- 8

A

B

C

D
- 9

A

B

C

D
- 10

A

B

C

D
- 11

A

B

C

D
- 12

A

B

C

D
- 13

A

B

C

D
- 14

A

B

C

D
- 15

A

B

C

D
- 16

A

B

C

D
- 17

A

B

C

D
- 18

A

B

C

D
- 19

A

B

C

D
- 20

A

B

C

D
- 21

A

B

C

D
- 22

A

B

C

D
- 23

A

B

C

D
- 24

A

B

C

D
- 25

A

B

C

D
- 26

A

B

C

D
- 27

A

B

C

D
- 28

A

B

C

D
- 29

A

B

C

D
- 30

A

B

C

D

Part 2: Mathematics Knowledge

- 1

A

B

C

D
- 2

A

B

C

D
- 3

A

B

C

D
- 4

A

B

C

D
- 5

A

B

C

D
- 6

A

B

C

D
- 7

A

B

C

D
- 8

A

B

C

D
- 9

A

B

C

D
- 10

A

B

C

D
- 11

A

B

C

D
- 12

A

B

C

D
- 13

A

B

C

D
- 14

A

B

C

D
- 15

A

B

C

D
- 16

A

B

C

D
- 17

A

B

C

D
- 18

A

B

C

D
- 19

A

B

C

D
- 20

A

B

C

D
- 21

A

B

C

D
- 22

A

B

C

D
- 23

A

B

C

D
- 24

A

B

C

D
- 25

A

B

C

D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

1. Last year monthly rent for an apartment was \$1,000. This year it is \$200. What is the percent decrease?
A 20% B 40% C 70% D 80%
2. Last year the price of a backpack was \$70. This year it is \$63. What is the percent decrease?
A 7% B 10% C 12% D 90%
3. Last year a monthly phone bill was \$68. This year it is \$102. What is the percent increase?
A 30% B 34% C 50% D 150%
4. Last year the price of a used car was \$15,800. This year it is \$17,380. What is the percent increase?
A 10% B 14% C 110% D 1,580%
5. Last year monthly rent for an apartment was \$1,200. This year it is \$720. What is the percent decrease?
A 25% B 40% C 60% D 480%
6. Last year the cost of a yearly gym membership was \$260. This year it is \$208. What is the percent decrease?
A 10% B 16% C 20% D 25%
7. Last year monthly rent for an apartment was \$1,600. This year it is \$1,920. What is the percent increase?
A 12% B 16% C 20% D 320%
8. Last year the price of a used car was \$17,900. This year it is \$12,530. What is the percent decrease?
A 20% B 30% C 70% D 5,370%
9. Last year the price of a pair of boots was \$112. This year it is \$196. What is the percent increase?
A 35% B 75% C 84% D 175%
10. Last year the price of a used car was \$19,000. This year it is \$9,500. What is the percent decrease?
A 40% B 50% C 100% D 9,500%
11. Last year the cost of a yearly gym membership was \$236. This year it is \$413. What is the percent increase?
A 45% B 75% C 175% D 177%
12. Last year a monthly phone bill was \$112. This year it is \$28. What is the percent decrease?
A 75% B 84% C 115% D 300%
13. Last year the cost of a yearly gym membership was \$340. This year it is \$408. What is the percent increase?
A 10% B 14% C 20% D 68%
14. Last year monthly rent for an apartment was \$2,100. This year it is \$2,520. What is the percent increase?
A 16% B 18% C 20% D 420%
15. Last year the price of a used car was \$19,100. This year it is \$4,775. What is the percent decrease?
A 7.5% B 25% C 35% D 75%
16. Last year the price of a pair of boots was \$224. This year it is \$56. What is the percent decrease?
A 25% B 75% C 168% D 300%
17. A training battalion has 850 recruits. 30% of them are assigned to the morning shift, and $\frac{2}{3}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 17 B 85 C 150 D 170
18. Last year the cost of a yearly gym membership was \$692. This year it is \$173. What is the percent decrease?
A 75% B 115% C 300% D 519%

19. A store sold a power drill for \$128. It increased the price by 50%, then put the power drill on sale at 50% off the new price. What is the sale price?
A \$64 B \$96 C \$128 D \$192
20. A training battalion has 300 recruits. 40% of them are assigned to the morning shift, and $\frac{1}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 20 B 22 C 24 D 120
21. A training battalion has 600 recruits. 30% of them are assigned to the morning shift, and $\frac{2}{3}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 12 B 60 C 120 D 180
22. A store sold a pair of running shoes for \$120. It increased the price by 50%, then put the pair of running shoes on sale at 50% off the new price. What is the sale price?
A \$9 B \$50 C \$60 D \$90
23. Last year the price of a gallon of paint was \$36. This year it is \$63. What is the percent increase?
A 7.5% B 27% C 37.5% D 75%
24. Last year the price of a tool kit was \$232. This year it is \$174. What is the percent decrease?
A 25% B 35% C 58% D 75%
25. A winter coat was priced at \$112. The store raised the price by 25%, and a month later it marked the new price down by 40%. What is the final price?
A \$84.00 B \$95.20 C \$104.00 D \$140.00
26. Seaman Kim surveyed 350 service members. 80% said they exercise every day, and $\frac{3}{5}$ of the daily exercisers said they run. How many said they run?
A 168 B 210 C 280 D 336
27. A store sold a power drill for \$132. It increased the price by 50%, then put the power drill on sale at 50% off the new price. What is the sale price?
A \$66 B \$99 C \$132 D \$198
28. A power drill was priced at \$96. The store raised the price by 50%, and a month later it marked the new price down by 50%. What is the final price?
A \$72 B \$96 C \$144 D \$720
29. A store sold a pair of running shoes for \$96. It increased the price by 50%, then put the pair of running shoes on sale at 50% off the new price. What is the sale price?
A \$36 B \$48 C \$52 D \$72
30. A training battalion has 550 recruits. 60% of them are assigned to the morning shift, and $\frac{2}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 132 B 198 C 220 D 330

Part 2: Mathematics Knowledge

Score: ____ / 25

Target time: 24 minutes

1. 72 is what percent of 120?
A 6% B 60% C 65% D 75%
2. 57 is what percent of 228?
A 25% B 45% C 50% D 400%
3. What is $\frac{1}{8}$ written as a percent?
A 1% B 1.25% C 9% D 12.5%
4. If 60% of a number is 261, what is the number?
A 321 B 435 C 870 D 15,660
5. If 10% of a number is 3, what is the number?
A 3 B 30 C 35 D 60
6. 14 is 10% of what number?
A 14 B 24 C 120 D 140
7. What is 120% of 700?
A 840 B 940 C 1,540 D 8,400
8. If 50% of a number is 170, what is the number?
A 34 B 85 C 220 D 340
9. What is $\frac{7}{8}$ written as a percent?
A 7% B 8.75% C 15% D 87.5%
10. What is $\frac{3}{8}$ written as a percent?
A 3% B 3.75% C 11% D 37.5%
11. Which of the following is equal to 0.93?
A 93% B 113% C 186% D 930%
12. What is 50% of 164?
A 52 B 62 C 82 D 820
13. 40 is 40% of what number?
A 10 B 16 C 80 D 100
14. What percent of 40 is 10?
A 0.25% B 2.5% C 25% D 400%
15. 56 is 40% of what number?
A 14 B 140 C 160 D 2,240
16. What is 50% of 84?
A 21 B 32 C 42 D 420
17. What is $\frac{3}{4}$ written as a percent?
A 3% B 75% C 105% D 750%
18. What is 25% of 40% of 480?
A 24 B 33 C 48 D 312
19. What is 75% of 60% of 240?
A 36 B 54 C 68 D 108
20. What is 40% of 50% of 500?
A 50 B 70 C 100 D 450
21. What is 75% of 60% of 800?
A 120 B 360 C 480 D 1,080
22. What is 50% of 20% of 600?
A 60 B 120 C 300 D 420
23. What is 25% of 75% of 720?
A 135 B 180 C 540 D 720
24. What is 75% of 60% of 200?
A 90 B 120 C 150 D 270
25. What is 20% of 75% of 800?
A 60 B 110 C 120 D 760

Answer Key: Practice Test 3

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 D ch. 6	7 C ch. 6	13 C ch. 6	19 B ch. 6	25 A ch. 6
2 B ch. 6	8 B ch. 6	14 C ch. 6	20 C ch. 6	26 A ch. 6
3 C ch. 6	9 B ch. 6	15 D ch. 6	21 C ch. 6	27 B ch. 6
4 A ch. 6	10 B ch. 6	16 B ch. 6	22 D ch. 6	28 A ch. 6
5 B ch. 6	11 B ch. 6	17 D ch. 6	23 D ch. 6	29 D ch. 6
6 C ch. 6	12 A ch. 6	18 A ch. 6	24 A ch. 6	30 A ch. 6

Part 2: Mathematics Knowledge

1 B ch. 6	6 D ch. 6	11 A ch. 6	16 C ch. 6	21 B ch. 6
2 A ch. 6	7 A ch. 6	12 C ch. 6	17 B ch. 6	22 A ch. 6
3 D ch. 6	8 D ch. 6	13 D ch. 6	18 C ch. 6	23 A ch. 6
4 B ch. 6	9 D ch. 6	14 C ch. 6	19 D ch. 6	24 A ch. 6
5 B ch. 6	10 D ch. 6	15 B ch. 6	20 C ch. 6	25 C ch. 6

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **D** 80%

- Find the amount of decrease: $1,000 - 200 = 800$ dollars.
- Divide by the *original* price: $\frac{800}{1,000} = 0.8$.
- Convert to a percent: $0.8 = 80\%$ decrease.

WHY NOT (A) gives the new price as a percent of the old price.

2. **B** 10%

- Find the amount of decrease: $70 - 63 = 7$ dollars.
- Divide by the *original* price: $\frac{7}{70} = 0.1$.
- Convert to a percent: $0.1 = 10\%$ decrease.

WHY NOT (A) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

3. **C** 50%

- Find the amount of increase: $102 - 68 = 34$ dollars.
- Divide by the *original* price: $\frac{34}{68} = 0.5$.
- Convert to a percent: $0.5 = 50\%$ increase.

WHY NOT (B) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

4. **A** 10%

- Find the amount of increase: $17,380 - 15,800 = 1,580$ dollars.
- Divide by the *original* price: $\frac{1,580}{15,800} = 0.1$.
- Convert to a percent: $0.1 = 10\%$ increase.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

5. **B** 40%

- Find the amount of decrease: $1,200 - 720 = 480$ dollars.
- Divide by the *original* price: $\frac{480}{1,200} = 0.4$.
- Convert to a percent: $0.4 = 40\%$ decrease.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

6. **C** 20%

- Find the amount of decrease: $260 - 208 = 52$ dollars.
- Divide by the *original* price: $\frac{52}{260} = 0.2$.
- Convert to a percent: $0.2 = 20\%$ decrease.

WHY NOT (D) divides by the new price instead of the original price.

7. **C** 20%

- Find the amount of increase: $1,920 - 1,600 = 320$ dollars.
- Divide by the *original* price: $\frac{320}{1,600} = 0.2$.
- Convert to a percent: $0.2 = 20\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

8. **B** 30%

- Find the amount of decrease: $17,900 - 12,530 = 5,370$ dollars.
- Divide by the *original* price: $\frac{5,370}{17,900} = 0.3$.
- Convert to a percent: $0.3 = 30\%$ decrease.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

9. **B** 75%

- Find the amount of increase: $196 - 112 = 84$ dollars.
- Divide by the *original* price: $\frac{84}{112} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

10. **B** 50%

- Find the amount of decrease: $19,000 - 9,500 = 9,500$ dollars.
- Divide by the *original* price: $\frac{9,500}{19,000} = 0.5$.
- Convert to a percent: $0.5 = 50\%$ decrease.

WHY NOT (C) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.

11. **B** 75%

- Find the amount of increase: $413 - 236 = 177$ dollars.
- Divide by the *original* price: $\frac{177}{236} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

12. **A** 75%

- Find the amount of decrease: $112 - 28 = 84$ dollars.
- Divide by the *original* price: $\frac{84}{112} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ decrease.

WHY NOT (B) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.

13. **C** 20%

- Find the amount of increase: $408 - 340 = 68$ dollars.
- Divide by the *original* price: $\frac{68}{340} = 0.2$.
- Convert to a percent: $0.2 = 20\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

14. **C** 20%

- Find the amount of increase: $2,520 - 2,100 = 420$ dollars.
- Divide by the *original* price: $\frac{420}{2,100} = 0.2$.
- Convert to a percent: $0.2 = 20\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

15. **D** 75%

- Find the amount of decrease: $19,100 - 4,775 = 14,325$ dollars.
- Divide by the *original* price: $\frac{14,325}{19,100} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ decrease.

WHY NOT (B) gives the new price as a percent of the old price.

16. **B** 75%

- Find the amount of decrease: $224 - 56 = 168$ dollars.
- Divide by the *original* price: $\frac{168}{224} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ decrease.

WHY NOT (A) gives the new price as a percent of the old price. (C) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.

17. **D** 170

- Find the first group: $0.3 \times 850 = 255$.
- Take $\frac{2}{3}$ of that group: $\frac{2}{3} \times 255 = 170$.

WHY NOT (B) is the part of the group that is *not* included.

18. **A** 75%

- Find the amount of decrease: $692 - 173 = 519$ dollars.
- Divide by the *original* price: $\frac{519}{692} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ decrease.

WHY NOT (C) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.

19. **B** \$96

- Raise the price by 50%: $128 \times 1.5 = 192$.
- Take 50% off the *new* price: $192 \times 0.5 = 96$.
- The final price is \$96.

WHY NOT (A) applies only the decrease. (C) assumes the two changes cancel out. (D) stops after the increase.

20. **C** 24

- Find the first group: $0.4 \times 300 = 120$.
- Take $\frac{1}{5}$ of that group: $\frac{1}{5} \times 120 = 24$.

WHY NOT (D) is the size of the first group only.

21. **C** 120

- Find the first group: $0.3 \times 600 = 180$.
- Take $\frac{2}{3}$ of that group: $\frac{2}{3} \times 180 = 120$.

WHY NOT (B) is the part of the group that is *not* included. (D) is the size of the first group only.

22. **D** \$90

- Raise the price by 50%: $120 \times 1.5 = 180$.
- Take 50% off the *new* price: $180 \times 0.5 = 90$.
- The final price is \$90.

WHY NOT (C) applies only the decrease.

23. **D** 75%

- Find the amount of increase: $63 - 36 = 27$ dollars.
- Divide by the *original* price: $\frac{27}{36} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (B) gives the dollar change instead of the percent change.

24. **A** 25%

- Find the amount of decrease: $232 - 174 = 58$ dollars.
- Divide by the *original* price: $\frac{58}{232} = 0.25$.
- Convert to a percent: $0.25 = 25\%$ decrease.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

25. **A** \$84.00

- Raise the price by 25%: $112 \times 1.25 = 140$.
- Take 40% off the *new* price: $140 \times 0.6 = 84$.
- The final price is \$84.

WHY NOT (B) combines the changes into a single 15% decrease. (D) stops after the increase.

26. **A** 168

- Find the first group: $0.8 \times 350 = 280$.
- Take $\frac{3}{5}$ of that group: $\frac{3}{5} \times 280 = 168$.

WHY NOT (B) takes $\frac{3}{5}$ of the whole total. (C) is the size of the first group only.

27. **B** \$99

- Raise the price by 50%: $132 \times 1.5 = 198$.
- Take 50% off the *new* price: $198 \times 0.5 = 99$.
- The final price is \$99.

WHY NOT (A) applies only the decrease. (C) assumes the two changes cancel out. (D) stops after the increase.

28. **A** \$72

- Raise the price by 50%: $96 \times 1.5 = 144$.
- Take 50% off the *new* price: $144 \times 0.5 = 72$.
- The final price is \$72.

WHY NOT (B) assumes the two changes cancel out. (C) stops after the increase.

29. **D** \$72

- Raise the price by 50%: $96 \times 1.5 = 144$.
- Take 50% off the *new* price: $144 \times 0.5 = 72$.
- The final price is \$72.

WHY NOT (B) applies only the decrease.

30. **A** 132

- Find the first group: $0.6 \times 550 = 330$.
- Take $\frac{2}{5}$ of that group: $\frac{2}{5} \times 330 = 132$.

WHY NOT (B) is the part of the group that is *not* included. (C) takes $\frac{2}{5}$ of the whole total. (D) is the size of the first group only.

Explanations: Mathematics Knowledge

1. **B** 60%
 - 1 Write the part over the whole: $\frac{72}{120}$.
 - 2 Simplify and convert: $\frac{72}{120} = 0.6 = 60\%$.

WHY NOT (A) moves the decimal point only one place.
2. **A** 25%
 - 1 Write the part over the whole: $\frac{57}{228}$.
 - 2 Simplify and convert: $\frac{57}{228} = 0.25 = 25\%$.

WHY NOT (D) divides the whole by the part instead of the part by the whole.
3. **D** 12.5%
 - 1 Divide the numerator by the denominator: $1 \div 8 = 0.125$.
 - 2 Multiply by 100 to get a percent: $0.125 = 12.5\%$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place.
4. **B** 435
 - 1 Use part = percent $\times$ whole: $261 = 0.6 \times w$.
 - 2 Divide both sides by 0.6: $w = 261 \div 0.6 = 435$.
 - 3 Check: 60% of 435 is 261. $\checkmark$

WHY NOT (A) adds the percent to the part. (D) multiplies by the percent without converting it.
5. **B** 30
 - 1 Use part = percent $\times$ whole: $3 = 0.1 \times w$.
 - 2 Divide both sides by 0.1: $w = 3 \div 0.1 = 30$.
 - 3 Check: 10% of 30 is 3. $\checkmark$

WHY NOT (A) slips the decimal point.
6. **D** 140
 - 1 Use part = percent $\times$ whole: $14 = 0.1 \times w$.
 - 2 Divide both sides by 0.1: $w = 14 \div 0.1 = 140$.
 - 3 Check: 10% of 140 is 14. $\checkmark$

WHY NOT (A) slips the decimal point. (B) adds the percent to the part.
7. **A** 840
 - 1 Change the percent to a decimal: $120\% = 1.2$.
 - 2 Multiply by the number: $1.2 \times 700 = 840$.

WHY NOT (C) adds 120% to the number. (D) moves the decimal point only one place.
8. **D** 340
 - 1 Use part = percent $\times$ whole: $170 = 0.5 \times w$.
 - 2 Divide both sides by 0.5: $w = 170 \div 0.5 = 340$.
 - 3 Check: 50% of 340 is 170. $\checkmark$

WHY NOT (A) slips the decimal point. (B) finds 50% of 170 instead. (C) adds the percent to the part.
9. **D** 87.5%
 - 1 Divide the numerator by the denominator: $7 \div 8 = 0.875$.
 - 2 Multiply by 100 to get a percent: $0.875 = 87.5\%$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place.
10. **D** 37.5%
 - 1 Divide the numerator by the denominator: $3 \div 8 = 0.375$.
 - 2 Multiply by 100 to get a percent: $0.375 = 37.5\%$.

WHY NOT (A) just writes the numerator as if it were a percent. (B) moves the decimal point only one place.
11. **A** 93%
 - 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
 - 2 $0.93 \times 100 = 93$, so $0.93 = 93\%$.

WHY NOT (D) moves the decimal point three places.
12. **C** 82
 - 1 Change the percent to a decimal: $50\% = 0.5$.
 - 2 Multiply by the number: $0.5 \times 164 = 82$.

WHY NOT (D) moves the decimal point only one place.
13. **D** 100
 - 1 Use part = percent $\times$ whole: $40 = 0.4 \times w$.
 - 2 Divide both sides by 0.4: $w = 40 \div 0.4 = 100$.
 - 3 Check: 40% of 100 is 40. $\checkmark$

WHY NOT (A) slips the decimal point. (B) finds 40% of 40 instead. (C) adds the percent to the part.
14. **C** 25%
 - 1 Write the part over the whole: $\frac{10}{40}$.
 - 2 Simplify and convert: $\frac{10}{40} = 0.25 = 25\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place. (D) divides the whole by the part instead of the part by the whole.
15. **B** 140
 - 1 Use part = percent $\times$ whole: $56 = 0.4 \times w$.
 - 2 Divide both sides by 0.4: $w = 56 \div 0.4 = 140$.
 - 3 Check: 40% of 140 is 56. $\checkmark$

WHY NOT (A) slips the decimal point. (D) multiplies by the percent without converting it.
16. **C** 42
 - 1 Change the percent to a decimal: $50\% = 0.5$.
 - 2 Multiply by the number: $0.5 \times 84 = 42$.

WHY NOT (D) moves the decimal point only one place.

17. **B** 75%

1 Divide the numerator by the denominator: $3 \div 4 = 0.75$.

2 Multiply by 100 to get a percent: $0.75 = 75\%$.

TIP Or scale the fraction to hundredths: $\frac{3}{4} = \frac{75}{100}$.

WHY NOT (A) just writes the numerator as if it were a percent.

18. **C** 48

1 Work from the inside out. First, 40% of 480: $0.4 \times 480 = 192$.

2 Then 25% of that result: $0.25 \times 192 = 48$.

TIP Percents of percents multiply: $0.25 \times 0.4 = 0.1$, and $0.1 \times 480 = 48$.

WHY NOT (D) adds the two percents.

19. **D** 108

1 Work from the inside out. First, 60% of 240: $0.6 \times 240 = 144$.

2 Then 75% of that result: $0.75 \times 144 = 108$.

TIP Percents of percents multiply: $0.75 \times 0.6 = 0.45$, and $0.45 \times 240 = 108$.

WHY NOT (A) subtracts the two percents.

20. **C** 100

1 Work from the inside out. First, 50% of 500: $0.5 \times 500 = 250$.

2 Then 40% of that result: $0.4 \times 250 = 100$.

TIP Percents of percents multiply: $0.4 \times 0.5 = 0.2$, and $0.2 \times 500 = 100$.

WHY NOT (A) subtracts the two percents. (D) adds the two percents.

21. **B** 360

1 Work from the inside out. First, 60% of 800: $0.6 \times 800 = 480$.

2 Then 75% of that result: $0.75 \times 480 = 360$.

TIP Percents of percents multiply: $0.75 \times 0.6 = 0.45$, and $0.45 \times 800 = 360$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 60%. (D) adds the two percents.

22. **A** 60

1 Work from the inside out. First, 20% of 600: $0.2 \times 600 = 120$.

2 Then 50% of that result: $0.5 \times 120 = 60$.

TIP Percents of percents multiply: $0.5 \times 0.2 = 0.1$, and $0.1 \times 600 = 60$.

WHY NOT (B) stops after taking 20%. (C) takes only 50% of 600. (D) adds the two percents.

23. **A** 135

1 Work from the inside out. First, 75% of 720: $0.75 \times 720 = 540$.

2 Then 25% of that result: $0.25 \times 540 = 135$.

WHY NOT (B) takes only 25% of 720. (C) stops after taking 75%. (D) adds the two percents.

24. **A** 90

1 Work from the inside out. First, 60% of 200: $0.6 \times 200 = 120$.

2 Then 75% of that result: $0.75 \times 120 = 90$.

TIP Percents of percents multiply: $0.75 \times 0.6 = 0.45$, and $0.45 \times 200 = 90$.

WHY NOT (B) stops after taking 60%. (C) takes only 75% of 200. (D) adds the two percents.

25. **C** 120

1 Work from the inside out. First, 75% of 800: $0.75 \times 800 = 600$.

2 Then 20% of that result: $0.2 \times 600 = 120$.

TIP Percents of percents multiply: $0.2 \times 0.75 = 0.15$, and $0.15 \times 800 = 120$.

WHY NOT (D) adds the two percents.

Practice Test 4

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

1. Last year monthly rent for an apartment was \$1,100. This year it is \$1,925. What is the percent increase?
A 75% B 150% C 175% D 825%
2. Last year the price of a tool kit was \$172. This year it is \$301. What is the percent increase?
A 37.5% B 75% C 129% D 175%
3. Last year the price of a pair of boots was \$164. This year it is \$123. What is the percent decrease?
A 25% B 41% C 75% D 250%
4. Last year the price of a pair of boots was \$224. This year it is \$336. What is the percent increase?
A 50% B 55% C 112% D 150%
5. Last year the price of a pair of boots was \$116. This year it is \$29. What is the percent decrease?
A 25% B 75% C 87% D 300%
6. Last year monthly rent for an apartment was \$1,700. This year it is \$1,275. What is the percent decrease?
A 25% B 45% C 75% D 425%
7. Last year the price of a used car was \$10,800. This year it is \$14,040. What is the percent increase?
A 30% B 35% C 130% D 3,240%
8. Last year monthly rent for an apartment was \$800. This year it is \$160. What is the percent decrease?
A 20% B 80% C 400% D 640%
9. Last year a monthly phone bill was \$120. This year it is \$144. What is the percent increase?
A 20% B 24% C 28% D 120%
10. Last year the price of a gallon of paint was \$56. This year it is \$14. What is the percent decrease?
A 25% B 42% C 75% D 300%
11. A warehouse holds 800 crates. 60% of the crates contain medical supplies, and $\frac{4}{5}$ of those crates must be kept refrigerated. How many crates must be refrigerated?
A 96 B 384 C 480 D 640
12. A winter coat was priced at \$184. The store raised the price by 25%, and a month later it marked the new price down by 10%. What is the final price?
A \$20.70 B \$165.60 C \$187.00 D \$207.00
13. Last year the price of a gallon of paint was \$20. This year it is \$15. What is the percent decrease?
A 5% B 15% C 20% D 25%
14. A training battalion has 800 recruits. 30% of them are assigned to the morning shift, and $\frac{4}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 48 B 132 C 152 D 192
15. A television was priced at \$224. The store raised the price by 25%, and a month later it marked the new price down by 25%. What is the final price?
A \$168 B \$210 C \$224 D \$280
16. Last year the price of a gallon of paint was \$24. This year it is \$18. What is the percent decrease?
A 6% B 25% C 30% D 75%
17. Last year the price of a used car was \$15,100. This year it is \$26,425. What is the percent increase?
A 7.5% B 35%
C 75% D 11,325%

18. Last year the price of a pair of boots was \$176. This year it is \$308. What is the percent increase?
A 7.5% B 37.5% C 75% D 132%
19. Last year the price of a tool kit was \$152. This year it is \$228. What is the percent increase?
A 35% B 50% C 76% D 150%
20. Last year the price of a used car was \$12,200. This year it is \$15,860. What is the percent increase?
A 3% B 15% C 30% D 3,660%
21. A store sold a power drill for \$164. It increased the price by 50%, then put the power drill on sale at 50% off the new price. What is the sale price?
A \$123 B \$164 C \$203 D \$246
22. A winter coat was priced at \$356. The store raised the price by 25%, and a month later it marked the new price down by 20%. What is the final price?
A \$156.00 B \$284.80 C \$356.00 D \$373.80
23. A training battalion has 700 recruits. 20% of them are assigned to the morning shift, and $\frac{1}{2}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 70 B 110 C 140 D 350
24. A training battalion has 950 recruits. 30% of them are assigned to the morning shift, and $\frac{1}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 42 B 47 C 57 D 285
25. A winter coat was priced at \$348. The store raised the price by 50%, and a month later it marked the new price down by 50%. What is the final price?
A \$174 B \$261 C \$348 D \$522
26. A training battalion has 100 recruits. 40% of them are assigned to the morning shift, and $\frac{3}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 16 B 24 C 40 D 60
27. A store sold a bicycle for \$696. It increased the price by 25%, then put the bicycle on sale at 50% off the new price. What is the sale price?
A \$435 B \$522 C \$870 D \$4,350
28. Seaman Kim surveyed 900 service members. 40% said they exercise every day, and $\frac{1}{2}$ of the daily exercisers said they run. How many said they run?
A 100 B 180 C 360 D 450
29. A store sold a power drill for \$72. It increased the price by 25%, then put the power drill on sale at 50% off the new price. What is the sale price?
A \$45 B \$54 C \$55 D \$90
30. A training battalion has 800 recruits. 45% of them are assigned to the morning shift, and $\frac{4}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 72 B 144 C 288 D 360

Part 2: Mathematics Knowledge

Score: ____ / 25

Target time: 24 minutes

1. What is 25% of 100?
A 5 B 15 C 25 D 250
2. 213 is what percent of 284?
A 0.75% B 7.5% C 75% D 150%
3. 222 is what percent of 148?
A 1.5% B 15% C 150% D 300%
4. What percent of 140 is 28?
A 0.2% B 2% C 16% D 20%
5. If 40% of a number is 98, what is the number?
A 138 B 145 C 195 D 245
6. What percent of 380 is 285?
A 0.75% B 7.5% C 75% D 150%
7. Which of the following is equal to 0.05?
A 0.05% B 0.5% C 4.5% D 5%
8. If 20% of a number is 42, what is the number?
A 210 B 420 C 840 D 2,100
9. Which of the following is equal to 0.79?
A 7.9% B 79% C 158% D 790%
10. 96 is 20% of what number?
A 480 B 630 C 960 D 1,920
11. What is $\frac{39}{40}$ written as a percent?
A 9.75% B 39% C 79% D 97.5%
12. If 80% of a number is 200, what is the number?
A 25 B 160 C 250 D 280
13. If 50% of a number is 225, what is the number?
A 45 B 450 C 4,500 D 11,250
14. What is 75% of 892?
A 669 B 869 C 1,561 D 6,690
15. What is 25% of 520?
A 13 B 50 C 65 D 130
16. What is $\frac{13}{20}$ written as a percent?
A 6.5% B 13% C 33% D 65%
17. 65 is 25% of what number?
A 26 B 90 C 160 D 260
18. What is 25% of 80% of 400?
A 60 B 80 C 320 D 420
19. What is 25% of 768?
A 96 B 132 C 192 D 1,920
20. What is 50% of 60% of 640?
A 64 B 192 C 384 D 704
21. What is 75% of 20% of 920?
A 69 B 118 C 138 D 874
22. What is 80% of 50% of 480?
A 144 B 192 C 240 D 624
23. What is 80% of 40% of 100?
A 27 B 32 C 40 D 120
24. What is 80% of 25% of 700?
A 14 B 100 C 140 D 735
25. What is 20% of 25% of 760?
A 38 B 152 C 190 D 342

Answer Key: Practice Test 4

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 A ch. 6	7 A ch. 6	13 D ch. 6	19 B ch. 6	25 B ch. 6
2 B ch. 6	8 B ch. 6	14 D ch. 6	20 C ch. 6	26 B ch. 6
3 A ch. 6	9 A ch. 6	15 B ch. 6	21 A ch. 6	27 A ch. 6
4 A ch. 6	10 C ch. 6	16 B ch. 6	22 C ch. 6	28 B ch. 6
5 B ch. 6	11 B ch. 6	17 C ch. 6	23 A ch. 6	29 A ch. 6
6 A ch. 6	12 D ch. 6	18 C ch. 6	24 C ch. 6	30 C ch. 6

Part 2: Mathematics Knowledge

1 C ch. 6	6 C ch. 6	11 D ch. 6	16 D ch. 6	21 C ch. 6
2 C ch. 6	7 D ch. 6	12 C ch. 6	17 D ch. 6	22 B ch. 6
3 C ch. 6	8 A ch. 6	13 B ch. 6	18 B ch. 6	23 B ch. 6
4 D ch. 6	9 B ch. 6	14 A ch. 6	19 C ch. 6	24 C ch. 6
5 D ch. 6	10 A ch. 6	15 D ch. 6	20 B ch. 6	25 A ch. 6

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **A** 75%

- Find the amount of increase: $1,925 - 1,100 = 825$ dollars.
- Divide by the *original* price: $\frac{825}{1,100} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

2. **B** 75%

- Find the amount of increase: $301 - 172 = 129$ dollars.
- Divide by the *original* price: $\frac{129}{172} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

3. **A** 25%

- Find the amount of decrease: $164 - 123 = 41$ dollars.
- Divide by the *original* price: $\frac{41}{164} = 0.25$.
- Convert to a percent: $0.25 = 25\%$ decrease.

WHY NOT (B) gives the dollar change instead of the percent change. (C) gives the new price as a percent of the old price.

4. **A** 50%

- Find the amount of increase: $336 - 224 = 112$ dollars.
- Divide by the *original* price: $\frac{112}{224} = 0.5$.
- Convert to a percent: $0.5 = 50\%$ increase.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

5. **B** 75%

- Find the amount of decrease: $116 - 29 = 87$ dollars.
- Divide by the *original* price: $\frac{87}{116} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ decrease.

WHY NOT (A) gives the new price as a percent of the old price. (C) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.

6. **A** 25%

- Find the amount of decrease: $1,700 - 1,275 = 425$ dollars.
- Divide by the *original* price: $\frac{425}{1,700} = 0.25$.
- Convert to a percent: $0.25 = 25\%$ decrease.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

7. **A** 30%

- Find the amount of increase: $14,040 - 10,800 = 3,240$ dollars.
- Divide by the *original* price: $\frac{3,240}{10,800} = 0.3$.
- Convert to a percent: $0.3 = 30\%$ increase.

WHY NOT (C) gives the new price as a percent of the old price. (D) gives the dollar change instead of the percent change.

8. **B** 80%

- Find the amount of decrease: $800 - 160 = 640$ dollars.
- Divide by the *original* price: $\frac{640}{800} = 0.8$.
- Convert to a percent: $0.8 = 80\%$ decrease.

WHY NOT (A) gives the new price as a percent of the old price. (C) divides by the new price instead of the original price. (D) gives the dollar change instead of the percent change.

9. **A** 20%

- Find the amount of increase: $144 - 120 = 24$ dollars.
- Divide by the *original* price: $\frac{24}{120} = 0.2$.
- Convert to a percent: $0.2 = 20\%$ increase.

WHY NOT (B) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

10. **C** 75%

- Find the amount of decrease: $56 - 14 = 42$ dollars.
- Divide by the *original* price: $\frac{42}{56} = 0.75$.
- Convert to a percent: $0.75 = 75\%$ decrease.

WHY NOT (A) gives the new price as a percent of the old price. (B) gives the dollar change instead of the percent change. (D) divides by the new price instead of the original price.

11. **B** 384

- Find the first group: $0.6 \times 800 = 480$.
- Take $\frac{4}{5}$ of that group: $\frac{4}{5} \times 480 = 384$.

WHY NOT (A) is the part of the group that is *not* included. (C) is the size of the first group only. (D) takes $\frac{4}{5}$ of the whole total.

12. **D** \$207.00

- Raise the price by 25%: $184 \times 1.25 = 230$.
- Take 10% off the *new* price: $230 \times 0.9 = 207$.
- The final price is \$207.

WHY NOT (B) applies only the decrease.

13. **D** 25%

- Find the amount of decrease: $20 - 15 = 5$ dollars.
- Divide by the *original* price: $\frac{5}{20} = 0.25$.
- Convert to a percent: $0.25 = 25\%$ decrease.

WHY NOT (A) gives the dollar change instead of the percent change.

14. **D** 192

- Find the first group: $0.3 \times 800 = 240$.
- Take $\frac{4}{5}$ of that group: $\frac{4}{5} \times 240 = 192$.

WHY NOT (A) is the part of the group that is *not* included.

15. **B** \$210

- 1 Raise the price by 25%: $224 \times 1.25 = 280$.
- 2 Take 25% off the *new* price: $280 \times 0.75 = 210$.
- 3 The final price is \$210.

WHY NOT (A) applies only the decrease. (C) assumes the two changes cancel out. (D) stops after the increase.

16. **B** 25%

- 1 Find the amount of decrease: $24 - 18 = 6$ dollars.
- 2 Divide by the *original* price: $\frac{6}{24} = 0.25$.
- 3 Convert to a percent: $0.25 = 25\%$ decrease.

WHY NOT (A) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

17. **C** 75%

- 1 Find the amount of increase: $26,425 - 15,100 = 11,325$ dollars.
- 2 Divide by the *original* price: $\frac{11,325}{15,100} = 0.75$.
- 3 Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

18. **C** 75%

- 1 Find the amount of increase: $308 - 176 = 132$ dollars.
- 2 Divide by the *original* price: $\frac{132}{176} = 0.75$.
- 3 Convert to a percent: $0.75 = 75\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

19. **B** 50%

- 1 Find the amount of increase: $228 - 152 = 76$ dollars.
- 2 Divide by the *original* price: $\frac{76}{152} = 0.5$.
- 3 Convert to a percent: $0.5 = 50\%$ increase.

WHY NOT (C) gives the dollar change instead of the percent change. (D) gives the new price as a percent of the old price.

20. **C** 30%

- 1 Find the amount of increase: $15,860 - 12,200 = 3,660$ dollars.
- 2 Divide by the *original* price: $\frac{3,660}{12,200} = 0.3$.
- 3 Convert to a percent: $0.3 = 30\%$ increase.

WHY NOT (D) gives the dollar change instead of the percent change.

21. **A** \$123

- 1 Raise the price by 50%: $164 \times 1.5 = 246$.
- 2 Take 50% off the *new* price: $246 \times 0.5 = 123$.
- 3 The final price is \$123.

WHY NOT (B) assumes the two changes cancel out. (D) stops after the increase.

22. **C** \$356.00

- 1 Raise the price by 25%: $356 \times 1.25 = 445$.
- 2 Take 20% off the *new* price: $445 \times 0.8 = 356$.
- 3 The final price is \$356.

WHY NOT (B) applies only the decrease. (D) combines the changes into a single 5% increase.

23. **A** 70

- 1 Find the first group: $0.2 \times 700 = 140$.
- 2 Take $\frac{1}{2}$ of that group: $\frac{1}{2} \times 140 = 70$.

WHY NOT (C) is the size of the first group only. (D) takes $\frac{1}{2}$ of the whole total.

24. **C** 57

- 1 Find the first group: $0.3 \times 950 = 285$.
- 2 Take $\frac{1}{5}$ of that group: $\frac{1}{5} \times 285 = 57$.

WHY NOT (D) is the size of the first group only.

25. **B** \$261

- 1 Raise the price by 50%: $348 \times 1.5 = 522$.
- 2 Take 50% off the *new* price: $522 \times 0.5 = 261$.
- 3 The final price is \$261.

WHY NOT (A) applies only the decrease. (C) assumes the two changes cancel out. (D) stops after the increase.

26. **B** 24

- 1 Find the first group: $0.4 \times 100 = 40$.
- 2 Take $\frac{3}{5}$ of that group: $\frac{3}{5} \times 40 = 24$.

WHY NOT (A) is the part of the group that is *not* included. (C) is the size of the first group only. (D) takes $\frac{3}{5}$ of the whole total.

27. **A** \$435

- 1 Raise the price by 25%: $696 \times 1.25 = 870$.
- 2 Take 50% off the *new* price: $870 \times 0.5 = 435$.
- 3 The final price is \$435.

WHY NOT (B) combines the changes into a single 25% decrease. (C) stops after the increase.

28. **B** 180

- 1 Find the first group: $0.4 \times 900 = 360$.
- 2 Take $\frac{1}{2}$ of that group: $\frac{1}{2} \times 360 = 180$.

WHY NOT (C) is the size of the first group only. (D) takes $\frac{1}{2}$ of the whole total.

29. **A** \$45

- 1 Raise the price by 25%: $72 \times 1.25 = 90$.
- 2 Take 50% off the *new* price: $90 \times 0.5 = 45$.
- 3 The final price is \$45.

WHY NOT (B) combines the changes into a single 25% decrease. (D) stops after the increase.

30. **C** 288

- 1 Find the first group: $0.45 \times 800 = 360$.
- 2 Take $\frac{4}{5}$ of that group: $\frac{4}{5} \times 360 = 288$.

WHY NOT (A) is the part of the group that is *not* included. (D) is the size of the first group only.

Explanations: Mathematics Knowledge

1. **C** 25

- 1 Change the percent to a decimal: $25\% = 0.25$.
- 2 Multiply by the number: $0.25 \times 100 = 25$.

WHY NOT (D) moves the decimal point only one place.

2. **C** 75%

- 1 Write the part over the whole: $\frac{213}{284}$.
- 2 Simplify and convert: $\frac{213}{284} = 0.75 = 75\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

3. **C** 150%

- 1 Write the part over the whole: $\frac{222}{148}$.
- 2 Simplify and convert: $\frac{222}{148} = 1.5 = 150\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

4. **D** 20%

- 1 Write the part over the whole: $\frac{28}{140}$.
- 2 Simplify and convert: $\frac{28}{140} = 0.2 = 20\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

5. **D** 245

- 1 Use part = percent $\times$ whole: $98 = 0.4 \times w$.
- 2 Divide both sides by 0.4: $w = 98 \div 0.4 = 245$.
- 3 Check: 40% of 245 is 98. ✓

WHY NOT (A) adds the percent to the part.

6. **C** 75%

- 1 Write the part over the whole: $\frac{285}{380}$.
- 2 Simplify and convert: $\frac{285}{380} = 0.75 = 75\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

7. **D** 5%

- 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- 2 $0.05 \times 100 = 5$, so $0.05 = 5\%$.

WHY NOT (A) forgets to move the decimal point at all. (B) moves the decimal point only one place.

8. **A** 210

- 1 Use part = percent $\times$ whole: $42 = 0.2 \times w$.
- 2 Divide both sides by 0.2: $w = 42 \div 0.2 = 210$.
- 3 Check: 20% of 210 is 42. ✓

WHY NOT (C) multiplies by the percent without converting it.

9. **B** 79%

- 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- 2 $0.79 \times 100 = 79$, so $0.79 = 79\%$.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.

10. **A** 480

- 1 Use part = percent $\times$ whole: $96 = 0.2 \times w$.
- 2 Divide both sides by 0.2: $w = 96 \div 0.2 = 480$.
- 3 Check: 20% of 480 is 96. ✓

WHY NOT (D) multiplies by the percent without converting it.

11. **D** 97.5%

- 1 Divide the numerator by the denominator: $39 \div 40 = 0.975$.
- 2 Multiply by 100 to get a percent: $0.975 = 97.5\%$.

WHY NOT (A) moves the decimal point only one place. (B) just writes the numerator as if it were a percent.

12. **C** 250

- 1 Use part = percent $\times$ whole: $200 = 0.8 \times w$.
- 2 Divide both sides by 0.8: $w = 200 \div 0.8 = 250$.
- 3 Check: 80% of 250 is 200. ✓

WHY NOT (A) slips the decimal point. (B) finds 80% of 200 instead. (D) adds the percent to the part.

13. **B** 450

- 1 Use part = percent $\times$ whole: $225 = 0.5 \times w$.
- 2 Divide both sides by 0.5: $w = 225 \div 0.5 = 450$.
- 3 Check: 50% of 450 is 225. ✓

WHY NOT (A) slips the decimal point. (D) multiplies by the percent without converting it.

14. **A** 669

- 1 Change the percent to a decimal: $75\% = 0.75$.
- 2 Multiply by the number: $0.75 \times 892 = 669$.

WHY NOT (C) adds 75% to the number. (D) moves the decimal point only one place.

15. **D** 130

- 1 Change the percent to a decimal: $25\% = 0.25$.
- 2 Multiply by the number: $0.25 \times 520 = 130$.

WHY NOT (A) moves the decimal point three places.

16. **D** 65%

- 1 Divide the numerator by the denominator: $13 \div 20 = 0.65$.
- 2 Multiply by 100 to get a percent: $0.65 = 65\%$.

TIP Or scale the fraction to hundredths: $\frac{13}{20} = \frac{65}{100}$.

WHY NOT (A) moves the decimal point only one place. (B) just writes the numerator as if it were a percent.

17. **D** 260

- 1 Use part = percent $\times$ whole: $65 = 0.25 \times w$.
- 2 Divide both sides by 0.25: $w = 65 \div 0.25 = 260$.
- 3 Check: 25% of 260 is 65. $\checkmark$

WHY NOT (A) slips the decimal point. (B) adds the percent to the part.

18. **B** 80

- 1 Work from the inside out. First, 80% of 400: $0.8 \times 400 = 320$.
- 2 Then 25% of that result: $0.25 \times 320 = 80$.

TIP Percents of percents multiply: $0.25 \times 0.8 = 0.2$, and $0.2 \times 400 = 80$.

WHY NOT (C) stops after taking 80%. (D) adds the two percents.

19. **C** 192

- 1 Change the percent to a decimal: $25\% = 0.25$.
- 2 Multiply by the number: $0.25 \times 768 = 192$.

WHY NOT (D) moves the decimal point only one place.

20. **B** 192

- 1 Work from the inside out. First, 60% of 640: $0.6 \times 640 = 384$.
- 2 Then 50% of that result: $0.5 \times 384 = 192$.

TIP Percents of percents multiply: $0.5 \times 0.6 = 0.3$, and $0.3 \times 640 = 192$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 60%. (D) adds the two percents.

21. **C** 138

- 1 Work from the inside out. First, 20% of 920: $0.2 \times 920 = 184$.

- 2 Then 75% of that result: $0.75 \times 184 = 138$.

TIP Percents of percents multiply: $0.75 \times 0.2 = 0.15$, and $0.15 \times 920 = 138$.

WHY NOT (D) adds the two percents.

22. **B** 192

- 1 Work from the inside out. First, 50% of 480: $0.5 \times 480 = 240$.

- 2 Then 80% of that result: $0.8 \times 240 = 192$.

TIP Percents of percents multiply: $0.8 \times 0.5 = 0.4$, and $0.4 \times 480 = 192$.

WHY NOT (A) subtracts the two percents. (C) stops after taking 50%. (D) adds the two percents.

23. **B** 32

- 1 Work from the inside out. First, 40% of 100: $0.4 \times 100 = 40$.

- 2 Then 80% of that result: $0.8 \times 40 = 32$.

TIP Percents of percents multiply: $0.8 \times 0.4 = 0.32$, and $0.32 \times 100 = 32$.

WHY NOT (C) stops after taking 40%. (D) adds the two percents.

24. **C** 140

- 1 Work from the inside out. First, 25% of 700: $0.25 \times 700 = 175$.

- 2 Then 80% of that result: $0.8 \times 175 = 140$.

TIP Percents of percents multiply: $0.8 \times 0.25 = 0.2$, and $0.2 \times 700 = 140$.

WHY NOT (D) adds the two percents.

25. **A** 38

- 1 Work from the inside out. First, 25% of 760: $0.25 \times 760 = 190$.

- 2 Then 20% of that result: $0.2 \times 190 = 38$.

TIP Percents of percents multiply: $0.2 \times 0.25 = 0.05$, and $0.05 \times 760 = 38$.

WHY NOT (B) takes only 20% of 760. (C) stops after taking 25%. (D) adds the two percents.

Formula Sheet

The ASVAB does not give you formulas. Everything on this page is fair game—learn it until you can write it from memory.

Percents & money

- part = percent $\times$ whole
- percent change = $\frac{\text{new} - \text{original}}{\text{original}} \times 100\%$
- sale price = price $\times$ (1 – discount rate)
- total with tax = price $\times$ (1 + tax rate)
- simple interest: $I = Prt$ (t in years)

Rates

- distance = rate $\times$ time, $d = rt$
- average speed = $\frac{\text{total distance}}{\text{total time}}$
- work together: $\frac{1}{a} + \frac{1}{b} = \frac{1}{t}$
- unit rate = $\frac{\text{amount}}{\text{one unit}}$

Statistics & probability

- mean = $\frac{\text{sum of values}}{\text{number of values}}$
- median: middle value of the *sorted* list
- mode: most frequent value; range = max – min
- $P(\text{event}) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$
- $P(\text{not } A) = 1 - P(A)$; independent: $P(A \text{ and } B) = P(A)P(B)$

Exponents & roots

- $a^m \cdot a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$
- $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$, $(ab)^n = a^n b^n$
- $\sqrt{ab} = \sqrt{a} \sqrt{b}$, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

Algebra

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$
- quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

- slope $m = \frac{y_2 - y_1}{x_2 - x_1}$; line: $y = mx + b$
- distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- midpoint = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Angles & triangles

- complementary: sum 90° ; supplementary: sum 180°
- angles of a triangle add to 180°
- polygon with n sides: angle sum $(n - 2) \times 180^\circ$
- Pythagorean theorem: $a^2 + b^2 = c^2$
- common triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25
- 45-45-90: sides $x, x, x\sqrt{2}$; 30-60-90: $x, x\sqrt{3}, 2x$

Perimeter & area

- rectangle: $P = 2l + 2w$, $A = lw$
- square: $P = 4s$, $A = s^2$
- triangle: $A = \frac{1}{2}bh$
- parallelogram: $A = bh$ (height, not slanted side)
- trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$

Circles & solids

- $d = 2r$, circumference $C = 2\pi r = \pi d$
- area $A = \pi r^2$; $\pi \approx 3.14 \approx \frac{22}{7}$
- box: $V = lwh$, $SA = 2lw + 2lh + 2wh$
- cube: $V = s^3$, $SA = 6s^2$
- cylinder: $V = \pi r^2 h$
- cone: $V = \frac{1}{3}\pi r^2 h$; sphere: $V = \frac{4}{3}\pi r^3$

Quick Reference Tables

Squares and cubes

$1^2 = 1$	$14^2 = 196$	$1^3 = 1$
$2^2 = 4$	$15^2 = 225$	$2^3 = 8$
$3^2 = 9$	$16^2 = 256$	$3^3 = 27$
$4^2 = 16$	$17^2 = 289$	$4^3 = 64$
$5^2 = 25$	$18^2 = 324$	$5^3 = 125$
$6^2 = 36$	$19^2 = 361$	$6^3 = 216$
$7^2 = 49$	$20^2 = 400$	$7^3 = 343$
$8^2 = 64$	$21^2 = 441$	$8^3 = 512$
$9^2 = 81$	$22^2 = 484$	$9^3 = 729$
$10^2 = 100$	$23^2 = 529$	$10^3 = 1,000$
$11^2 = 121$	$24^2 = 576$	
$12^2 = 144$	$25^2 = 625$	
$13^2 = 169$		

Fractions, decimals, percents

Fraction	Decimal	Percent
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	0.333...	$33\frac{1}{3}\%$
$\frac{2}{3}$	0.666...	$66\frac{2}{3}\%$
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{3}{5}$	0.6	60%
$\frac{4}{5}$	0.8	80%
$\frac{1}{6}$	0.1666...	$16\frac{2}{3}\%$
$\frac{1}{8}$	0.125	12.5%
$\frac{3}{8}$	0.375	37.5%
$\frac{5}{8}$	0.625	62.5%
$\frac{7}{8}$	0.875	87.5%
$\frac{1}{10}$	0.1	10%
$\frac{1}{20}$	0.05	5%
$\frac{1}{25}$	0.04	4%
$\frac{1}{50}$	0.02	2%
$\frac{1}{100}$	0.01	1%

Prime numbers less than 100

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

Divisibility rules

- 2 last digit is even
- 3 sum of digits divisible by 3
- 4 last two digits divisible by 4
- 5 last digit 0 or 5
- 6 divisible by 2 and by 3
- 9 sum of digits divisible by 9
- 10 last digit 0

Units of measure

12 inches = 1 foot; 3 feet = 1 yard
 5,280 feet = 1 mile; 1,760 yards = 1 mile
 8 fluid ounces = 1 cup; 2 cups = 1 pint
 2 pints = 1 quart; 4 quarts = 1 gallon
 16 ounces = 1 pound; 2,000 pounds = 1 ton
 60 seconds = 1 minute; 60 minutes = 1 hour
 $1 \text{ yard}^2 = 9 \text{ feet}^2$; $1 \text{ foot}^2 = 144 \text{ inches}^2$
 $1 \text{ yard}^3 = 27 \text{ feet}^3$

Metric: kilo = 1,000; centi = $\frac{1}{100}$; milli = $\frac{1}{1000}$
 1 km = 1,000 m; 1 m = 100 cm = 1,000 mm
 1 kg = 1,000 g; 1 L = 1,000 mL

24-hour (military) time

Hours after noon: add 12. 1:00 p.m. = 1300,
 6:45 p.m. = 1845, midnight = 0000, 12:30 a.m.
 = 0030.

Glossary of Math Terms

Mathematics Knowledge questions often test vocabulary directly (“Which of the following is a prime number?”). Make sure every term below is familiar.

Absolute value. The distance of a number from 0 on the number line; never negative. $|-7| = 7$.

Acute angle. An angle greater than 0° and less than 90° .

Area. The amount of surface a flat figure covers, measured in square units.

Average (mean). The sum of a set of values divided by the number of values.

Binomial. A polynomial with two terms, such as $x + 5$.

Circumference. The distance around a circle: $C = 2\pi r = \pi d$.

Coefficient. The number multiplying a variable. In $7x^2$, the coefficient is 7.

Complementary angles. Two angles whose measures add up to 90° .

Composite number. A whole number greater than 1 that has more than two factors, such as 9 or 12.

Congruent. Having exactly the same size and shape (equal sides, equal angles).

Consecutive integers. Integers that follow one another in order, such as 7, 8, 9.

Constant. A term with no variable, such as the 5 in $3x + 5$.

Denominator. The bottom number of a fraction; it tells how many equal parts make a whole.

Diameter. A segment through the center of a circle from side to side; twice the radius.

Difference. The result of a subtraction.

Digit. Any of the symbols 0–9 used to write numbers.

Divisor. The number you divide by. In $20 \div 4$, the divisor is 4.

Equation. A statement that two expressions are equal, such as $2x + 3 = 11$.

Equilateral triangle. A triangle with three equal sides and three 60° angles.

Even number. An integer divisible by 2.

Exponent. The small raised number that tells how many times the base is used as a factor: $2^3 = 2 \cdot 2 \cdot 2$.

Factor. A number that divides another number evenly. The factors of 12 are 1, 2, 3, 4, 6, 12.

Greatest common factor (GCF). The largest number that divides each of two or more numbers.

Hypotenuse. The longest side of a right triangle, opposite the right angle.

Improper fraction. A fraction whose numerator is greater than or equal to its denominator, such as $\frac{7}{4}$.

Inequality. A statement using $<$, $>$, $\leq$, or $\geq$ to compare two expressions.

Integer. A whole number or its opposite: $\dots, -2, -1, 0, 1, 2, \dots$

Irrational number. A number that cannot be written as a fraction of integers, such as $\sqrt{2}$ or π .

Isosceles triangle. A triangle with at least two equal sides; the angles opposite them are equal.

Least common multiple (LCM). The smallest positive number that is a multiple of each of two or more numbers.

Like terms. Terms with the same variables raised to the same powers, such as $3x^2$ and $-5x^2$.

Linear equation. An equation whose graph is a straight line; the variable has no exponent above 1.

Median. The middle value of a sorted list (or the average of the two middle values).

Mixed number. A whole number and a fraction together, such as $2\frac{1}{3}$.

Mode. The value that occurs most often in a data set.

Monomial. A polynomial with one term, such as $-4x^3y$.

Multiple. The product of a number and a whole number. Multiples of 6: 6, 12, 18, $\dots$

Numerator. The top number of a fraction.

Obtuse angle. An angle greater than 90° and less than 180° .

Odd number. An integer that is not divisible by 2.

Parallel lines. Lines in the same plane that never meet; they have equal slopes.

Percent. Per hundred: $35\% = \frac{35}{100} = 0.35$.

Perimeter. The distance around a figure.

Perpendicular lines. Lines that meet at a right angle; their slopes are negative reciprocals.

Polygon. A closed figure made of straight sides (triangle, quadrilateral, pentagon, $\dots$).

Polynomial. A sum of terms of the form ax^n , such as $3x^2 - 5x + 2$.

Prime number. A whole number greater than 1 whose only factors are 1 and itself: 2, 3, 5, 7, 11, ...

Probability. The chance that an event occurs, from 0 (impossible) to 1 (certain).

Product. The result of a multiplication.

Proportion. An equation stating that two ratios are equal, such as $\frac{3}{4} = \frac{9}{12}$.

Quadrant. One of the four regions of the coordinate plane, numbered I–IV counterclockwise from the upper right.

Quadratic equation. An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$.

Quotient. The result of a division.

Radius. The distance from the center of a circle to any point on it.

Range. The difference between the largest and smallest values in a data set.

Ratio. A comparison of two quantities by division, written $3 : 4$ or $\frac{3}{4}$.

Rational number. Any number that can be written as a fraction of two integers.

Reciprocal. The number that multiplies with a given number to give 1. The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

Right angle. An angle of exactly 90° .

Scientific notation. A number written as $a \times 10^n$ with $1 \leq a < 10$, such as 4.2×10^5 .

Slope. The steepness of a line: rise over run, $\frac{y_2 - y_1}{x_2 - x_1}$.

Square root. A number that, multiplied by itself, gives the original number: $\sqrt{49} = 7$.

Sum. The result of an addition.

Supplementary angles. Two angles whose measures add up to 180° .

Trinomial. A polynomial with three terms, such as $x^2 + 5x + 6$.

Variable. A letter that stands for an unknown or changing number.

Vertex. A corner point of an angle, polygon, or solid (plural: vertices).

Vertical angles. The opposite angles formed by two intersecting lines; they are equal.

Volume. The amount of space a solid occupies, measured in cubic units.

Whole numbers. The numbers 0, 1, 2, 3, ...

Progress Tracker

Record your score the first time you complete each practice set, then redo the problems you missed about a week later and record that score too.

Ch.	Practice set	Score	Date	Redo
6	Percents	/ 25		/ 25
–	Diagnostic Test	/ 30		
–	Practice Test 1	/ 55		
–	Practice Test 2	/ 55		
–	Practice Test 3	/ 55		
–	Practice Test 4	/ 55		

Thank you, and good luck!

If this workbook helped you prepare, please consider leaving a short review on Amazon. Reviews help other future service members find the right study materials—and they help the author keep improving this book.

Found an error, or a solution you think could be explained better? Every report is read and fixed in the next printing.